Notes
Biostatistical Methods I

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0.1 Introduction

The notes in this PDF are based on the course "Biostatistical Methods I" taught by Dr. Guha at UF in the Fall semester of 2023. The course, and these subsequent notes, utilizes the textbook *"Bayesian and Frequentist Regression Methods" by Jon Wakefield* and contains complete exerts from the book. This book can be found online.
A word of caution, these are notes and not a textbook. These are under the edit of no professor and should be considered as trustworthy as a blog post. In addition, the textbook provided contains many typos and by correlation there are many typos in these notes and it should be noted that if you do not confirm the equations yourself (which you should for practice) then the book cannot be used effectively.

0.1.1 Overview

The notes will cover two statistical frameworks; Bayesian and Frequentist. A general outline of the ideas will be provided along with surface level explanations of the statistical methods, estimation procedures, and sampling models. Full mastery of the topics cannot be gained from studying these notes, but some naive examples and proofs will be elaborated upon. A full description of home works (in order) from the course is also provided within the text once sufficient material has been covered to complete them.

1 Frequentist Inference

1.0.1 Introduction to Frequentist Framework

Under the frequentist framework, parameters and hypotheses are viewed as unknown, but fixed (nonrandom), quantities, and consequently there is no possibility of making probability statements about these unknowns. Frequentist approaches are often generalized and need to be "fixed" for various situations. The use of asymptotics and limits to describe what *should happen* to a statistic is the basis of frequentist ideas.

1.0.2 Frequentist Criteria

Consider the univariate parameter θ and let $\mathbf{Y} = [Y_1, \ldots, Y_n]^T$ be a set of random variables with realizations $\mathbf{y} = [y_1, \ldots, y_n]^T$. We often represent our inference with use of a 100(1-α)% *confidence interval* for θ, such as

$$[a(\mathbf{Y}), b(\mathbf{Y})] \text{ where } 1 - \alpha = Pr(\theta \in [a(\mathbf{Y}), b(\mathbf{Y})]).$$

Note that a and b depend on the random variables in $\mathbf{Y}$ rather than the realization as we cannot make inference on the realizations. Ideally we want the shortest confidence interval for a given α, this will be the main criteria of how we determine the best estimator. To measure the closeness of the data to the centered point estimator $\hat{\theta}(\mathbf{Y})$, we have the *mean squared error* which is described as "the expected, squared difference between the parameter estimate and its true value"

$$
\begin{aligned}
MSE[\hat{\theta}(Y)] &= E_{\mathbf{Y}|\theta}\left[(\hat{\theta}(\mathbf{Y})-\theta)^2\right] \\
&= E_{\mathbf{Y}|\theta}\left[\hat{\theta}(\mathbf{Y})^2\right] + E_{\mathbf{Y}|\theta}\left[\theta^2\right] - 2E_{\mathbf{Y}|\theta}\left[\hat{\theta}(\mathbf{Y})\theta\right] \\
&= \underbrace{E_{\mathbf{Y}|\theta}\left[\hat{\theta}(\mathbf{Y})^2\right] - E_{\mathbf{Y}|\theta}\left[\hat{\theta}(\mathbf{Y})\right]^2}_{\text{variance equation}} + \underbrace{\theta^2 - 2\theta E_{\mathbf{Y}|\theta}\left[\hat{\theta}(\mathbf{Y})\right] + E_{\mathbf{Y}|\theta}\left[\hat{\theta}(\mathbf{Y})\right]^2}_{\text{theta constant by frequentist ideals (above)}} \\
&= \underbrace{\mathrm{Var}_{\mathbf{Y}|\theta}\left[\hat{\theta}(\mathbf{Y})\right]}_{\text{variance of }\hat{\theta}} + \underbrace{\left(E_{\mathbf{Y}|\theta}\left[\hat{\theta}(\mathbf{Y})\right]-\theta\right)^2}_{\text{bias of }\hat{\theta}} \\
&= \mathrm{Var}_{\mathbf{Y}|\theta}\left[\hat{\theta}(\mathbf{Y})\right] + \text{bias}\left[\hat{\theta}(\mathbf{Y})\right]^2.
\end{aligned}
$$

The notation stresses that all expectations are with respect to the sampling distribution of the estimator given the true value of the parameter. In the future this notation will be suppressed (i.e. $E_{\mathbf{Y}|\theta} \implies E$).

Keep in mind the motivation going forward is to **lower the variance and remove bias**, thus resulting in more precise and accurate inference, respectively. First we must introduce the tools in which we will do this.

1.1 Sufficient Statistics

Let $\mathbf{Y} = [Y_1, \ldots, Y_n]^T$ be a random sample from the density $p(\mathbf{Y}|\theta)$. We say the statistic $T(\mathbf{Y})$ is *sufficient* for θ within a family of probability distributions, $p(\mathbf{Y}|\theta)$, if $P(\mathbf{Y}|T(\mathbf{Y}))$ does not depend upon θ. Furthermore, **Fisher-Neyman Factorization Theorem** states that $T(\boldsymbol{Y})$ *is a sufficient statistic iff* $p(\boldsymbol{Y}|\theta) = g[T(\boldsymbol{Y})|\theta] \times h(\boldsymbol{Y})$. The idea is that "all information about θ can be explained by $T(\mathbf{Y})$."

The statistic $T(\mathbf{Y})$ is *minimally sufficient* for θ within a family of probability distributions $p(\mathbf{Y}|\theta)$ if no further reductions from T is possible while retaining sufficiency. The **Lehmann-Scheffé Theorem** states that *if* $T(\boldsymbol{Y})$ *satisfies the following property: for every pair of sample points* $\boldsymbol{Y}, \boldsymbol{X}$ *we have that* $p(\boldsymbol{Y}|\theta)/p(\boldsymbol{X}|\theta)$ *is free from* θ *iff* $T(\boldsymbol{Y}) = T(\boldsymbol{X})$*, then* T *is minimally sufficient.*

Example 1 Let $Y_1, \ldots, Y_n$ be an iid sample from the exponential distribution. Say we want to estimate the mean of the distribution, so our focus is on the statistic λ.

$$
p(\mathbf{Y}|\lambda) = \prod_i^n \lambda \exp(-\lambda Y_i) = \underbrace{\exp\left(-\lambda \sum_i^n Y_i + \log(\lambda)\right) \times 1}_{g[T(\mathbf{Y})|\lambda]\times h(\mathbf{Y})}
$$

We can see that $\sum_i^n Y_i$ itself is a sufficient statistic as it passes the tests above (this make sense because we use sample mean to estimate mean). Given another sample $X_1, \ldots, X_n$ from the same distribution. Only if $\sum X_i = \sum Y_i$, we have the following,

$$
\frac{p(\mathbf{Y}|\lambda)}{p(\mathbf{X}|\lambda)} = \exp\left(\lambda \sum_i^n [X_i - Y_i]\right) = 1,
$$

which is free of λ. Therefore by *Lehman-Scheffé theorem* $\sum_i^n Y_i$ is minimal sufficient.

Example 2 Let $Y_1, \ldots, Y_n$ be an iid sample from a centered normal distribution. Say we want to estimate the variance σ^2.

$$p(\mathbf{Y}|\sigma^2) = \prod_i^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{Y_i^2}{2\sigma^2}\right) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_i^n Y_i^2\right)$$

In the same fashion as example 1, if we assume that another sample $X_1, \ldots, X_n$ is given from the same distribution. Only if $\sum X_i^2 = \sum Y_i^2$, we have

$$\frac{p(\mathbf{Y}|\sigma^2)}{p(\mathbf{X}|\sigma^2)} = 1,$$

which is free of σ^2. Therefore by *Lehman-Scheffé theorem* $\sum_i^n Y_i^2$ is minimal sufficient when the mean is known and zero.

Example 3 Let $Y_1, \ldots, Y_n$ be an iid sample from a normal distribution. Say we want to estimate the mean, μ, and variance is known.

$$\begin{aligned}
p(\mathbf{Y}|\mu) &= \prod_i^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(Y_i-\mu)^2}{2\sigma^2}\right) \\
&= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_i^n [Y_i^2 - 2\mu Y_i + \mu^2]\right) \\
&= \underbrace{(2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_i^n Y_i^2\right)}_{h(\mathbf{y})\text{ “doesn't contain }\mu\text{”}} \times \underbrace{\exp\left(\frac{\mu}{\sigma^2}\sum_i^n Y_i - \frac{n\mu^2}{2\sigma^2}\right)}_{g[T(\mathbf{Y})|\mu]}
\end{aligned}$$

Fisher-Neyman factorization theorem tells us that from the above the statistic $\sum_i^n Y_i$ is sufficient. Furthermore, it is very easy to use the *Lehman-Scheffé theorem* from this point, so the statistic is minimal sufficient.

1.1.1 Conditionally Sufficient and Ancillary Statistics

If a minimal statistic is $\mathbf{T} = [T_1, T_2]$ and T_2 is ancillary then T_1 is called *conditionally sufficient* given T_2. *A statistic $V = V(\theta)$ is ancillary for θ within a family of probability distributions $p(y|\theta)$ if its distribution does not depend on θ.*

note: in the context of regression, ancillary statistics can be thought of as covariates that help define $\mathbf{Y}$. For example, if we have some normal linear regression where the covariates $\mathbf{x} = [x_1, \ldots, x_n]^T$ have distribution $p(x)$ and

$$(Y_i|X_i = x_i) \sim N(\beta_0 + \beta_1 X_i, \sigma^2)$$

then letting $\mathbf{y} = [y_1, \ldots, y_n]^T$ and $\beta = [\beta_0, \beta_1]^T$ the distribution of the data is

$$p(\mathbf{x}, \mathbf{y}|\beta, \sigma^2) = p(\mathbf{x})(2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_i^n [y_i - \beta_0 - \beta_1 x_i]^2\right).$$

The sufficient statistic for $[\beta, \sigma^2]$ is

$$\mathbf{S} = \left[\hat{\beta}, \hat{\sigma}^2, \sum_{i=1}^{n} x_i, \sum_{i=1}^{n} x_i^2\right]$$

where the last two terms are ancillary statistics (this proof is left for homework).

1.1.2 Complete Statistic

A statistic T is said to be complete if for every measurable function $g(\cdot)$ the expectation $E[g(T)] = 0$ for every θ implies that $g(T) = 0$ for every θ. (in other words: given the setup that $E[g(T)] = 0$, we have that $g(T) = 0$, then T is complete)

Example 4 Consider the setup from example 1, we want to show that $T = \sum y_i$ is a complete statistic. So we must show that

$$E(g(T)) = \int_{y \in (0,\infty)} g(T)p(y|\lambda)dy = 0 \implies g(T) = 0$$

We start by looking at the distribution of T. Each y_i is iid and exponentially distributed. Since $T = \sum_i y_i$, we need the distribution of the sum of n many iid exponential distributed random variables. It can be shown (do this for practice, trust me it'll be useful later) that the sum of n many iid exponential distributions has distributions $\sum_i^n y_i \sim Gamma(n, 1/\lambda)$. Therefore we have

$$\begin{aligned} 0 = E(g(T)) &= \int_{y \in (0,\infty)} g(T) \frac{(\lambda)^n}{\Gamma(n)} y^{n-1} \exp(-y\lambda) dy \\ &= \frac{(\lambda)^n}{\Gamma(n)} \int_{y \in (0,\infty)} g(T) y^{n-1} \exp(-y\lambda) dy \\ \implies 0 &= \int_{y \in (0,\infty)} g(T) y^{n-1} \exp(-y\lambda) dy. \end{aligned}$$

The function of $y^{n-1}\exp(-y\lambda)$ is always greater than 0 on the domain of $(0, \infty)$ (a positive number raised to any power is always positive) so we may only have the area under its curve equal to zero when $g(T) = 0$. Therefore $T = \sum y_i$ is a complete statistic.

1.2 UMVUEs

We saw earlier that the goal with estimators is to reduce the mean squared error, which will close the confidence interval around the correct location. The ideal estimator would have bias equal to zero and the lowest possible variance. What we want is to find uniformly minimum-variance unbiased estimators (UMVUEs). It should be noted that this approach for finding an estimator is to first remove bias, then lower variance.

Rao-Blackwell-Lehman-Scheffé Theorem *Let $T = T(\boldsymbol{Y})$ be a complete and sufficient statistic for θ. If there exists at least one unbiased estimator $\tilde{\phi} = \tilde{\phi}(\boldsymbol{Y})$ for $\phi(\theta)$ then there exists a unique UMVUE $\hat{\phi} = \hat{\phi}(T)$ for $\phi(\theta)$, namely,*

$$\phi(\theta) = E[\tilde{\phi}(\mathbf{Y}|\mathbf{T})]$$

Corollary *Let $T = T(\boldsymbol{Y})$ be complete and sufficient for θ. Then any function $g(T)$ is the UMVUE of its expectation $E[g(T)] = \phi(\theta)$.*

Suppose we want to estimate $\phi = \phi(\theta)$ based on $y|\theta \sim p(\cdot|\theta)$. The theory depends first on writing down the full probability model and assuming conditional independence, that is

$$p(\boldsymbol{y}|\theta) = \prod_{i=1}^{n} p(y_i|\theta)$$

If we let $T(\mathbf{Y})$ be a sufficient statistic of dimension 1, then, under suitable regularity conditions, there is a unique function $\phi(\theta)$ for which a UMVUE exists and its variance attains the Cramér-Rao lower bound (A short derivation of the Cramér-Rao lower bound will be provided at the end of this section). It should be noted that this only holds when the data are independently sampled from a one-parameter exponential family. If the previous statement is held, the distribution can be written in the form

$$p(y|\theta) = \exp\left(\theta T(y) - b(\theta) + c(y)\right).$$

For an example of this, refer to the first example of sufficient statistics of the exponential distribution, you'll see this exact form with $c(y) = 0$).

Example 5 Consider a single observation from a Poisson distribution: $Y|\theta \sim \text{Poisson}(\theta)$. Suppose we wish to estimate $\exp(-3\theta)$. Show that the UMVUE is $(-2)^y$ for $y = 0, 1, 2, \ldots$ Is this a reasonable estimator?
Here we will begin with the observation that the Poisson distribution is from the exponential family.

$$p(y|\theta) = \frac{e^{-\theta}\theta^y}{y!} = \underbrace{\exp(\log(\theta)y - \theta - \log(y!))}_{\exp(\theta T(y) - b(\theta) + c(y))} \implies T(y) = y.$$

So we have that y is sufficient, furthermore it is minimally sufficient because it cannot be reduced. To show that it is a complete statistic we can use the definition of complete statistics from above,

$$0 = E(g(T)) = \sum_{y=0}^{\infty} g(T)\frac{e^{-\theta}\theta^y}{y!} \implies g(T) = 0.$$

Clearly this is true as the sum contains only positive values, thus can only be zero if for every iteration $g(T) = 0$.
Now that we have y as a complete and minimal sufficient statistic, from the corollary, any func-

tions of the complete and sufficient statistic is the UMVUE of its expectation, that is

$$\begin{aligned}
\phi(\theta) &= E(g(T)) \\
\implies \phi(\theta) &= E((-2)^y) \\
&= \sum_{y=0}^{\infty} (-2)^y \frac{e^{-\theta}\theta^y}{y!} \\
&= \sum_{y=0}^{\infty} \frac{e^{-\theta}(-2\theta)^y}{y!} \\
&= \exp(-3\theta) \sum_{y=0}^{\infty} \frac{e^{2\theta}(-2\theta)^y}{y!} \\
&= \exp(-3\theta)
\end{aligned}$$

So we have that $(-2)^y$ is the UMVUE for $\exp(-3\theta)$. However, this is a pretty bad estimator, as the estimator can take negative values (what it's estimating is always positive).

Cramér-Rao lower bound derivation for univariate estimator: The Cramér-Rao lower bound for any unbiased estimator $\hat{\phi}$ of a scalar function of interest $\phi = \phi(\theta)$,

$$\text{Var}(\hat{\phi}) \geq \frac{[\phi'(\theta)^2]}{E[\frac{\partial^2 l}{\partial \theta^2}]}$$

where l is the log-likelihood of the joint distribution $l(\theta) = \sum_{i=1}^{n} \log p(y_i|\theta)$.

proof: Assume we have a random variable Y with density $f(Y;\theta)$ with parameter θ and estimator $T = t(Y)$ such that $E(T) = \psi(\theta)$. Define S as the score function where

$$S = \frac{\partial}{\partial \theta} \ln f(Y;\theta) = \frac{1}{f(Y;\theta)} \frac{\partial}{\partial \theta} f(Y;\theta)$$

and

$$E(S) = \int \left[\frac{1}{f(y;\theta)} \frac{\partial}{\partial \theta} f(y;\theta) \right] f(y;\theta) dy = \frac{\partial}{\partial \theta} \int f(y;\theta) dy = 0$$

If we consider the covariance of T and S we can see that

$$\text{cov}(S,T) = E(ST) - E(S)E(T) = E(ST) - 0 \cdot \psi(\theta) = E(ST)$$

so we have that

$$\text{cov}(S,T) = E(ST) = \int t(y) \left[\frac{1}{f(y;\theta)} \frac{\partial}{\partial \theta} f(y;\theta) \right] f(y;\theta) dy = \frac{\partial}{\partial \theta} \int t(x) f(y;\theta) dy = \frac{\partial}{\partial \theta} E(t(x)) = \psi'(\theta)$$

The Cauchy-Schwarz inequality shows that

$$|\text{Cov}(S,T)|^2 \leq \text{Var}(S)\text{Var}(T)$$

which implies

$$\text{Var}(T) \geq \frac{\psi'(\theta)^2}{\text{Var}(S)} = \frac{\psi'(\theta)^2}{E(S^2)} = \frac{\psi'(\theta)^2}{E(\frac{\partial^2 l}{\partial \theta^2})}.$$

1.2.1 Recap

Up until now we've found a sufficient statistic and have shown that it converges to the parameter we want to estimate or rather that the estimator is consistent (see book appendix F). Once we've shown consistency we want asymptotic normality, with this we can conduct inference through the 100(1-α)% confidence interval about the predicted parameter,

$$\hat{\theta}_n \pm z_{1-\alpha/2} \cdot \sqrt{\text{Var}(\hat{\theta}_n)},$$

where $Z \sim N(0,1)$ and $Pr(Z < z_{1-\alpha/2}) = 1 - \alpha/2$. Recall once again, the goal is to shorten this confidence interval as much as possible while retaining validity of the estimator and its statistics.

1.3 Estimating Functions

Let $\mathbf{Y} = [Y_1, \ldots, Y_n]^T$ represent n iid observations from a distribution indexed by a p-dimensional parameter θ. An estimating function is a function of the form

$$G_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} G(\theta, Y_i) \quad \text{where} \quad E(G_n(\theta)) = 0 \quad \text{for all } \theta.$$

The corresponding estimating equation which defines the estimator $\hat{\theta}_n$ has the form and restriction

$$G_n(\hat{\theta}_n) = \frac{1}{n}\sum_{i=1}^{n} G(\hat{\theta}_n, Y_i) = 0.$$

We can utilize the Taylor series to get the asymptotic distribution. First, let's evaluate the estimating equation at the true value of the parameter θ,

$$0 = G_n(\hat{\theta}_n) = G_n(\theta) + (\hat{\theta}_n - \theta)G_n'(\tilde{\theta}).$$

where $\tilde{\theta}$ is between θ and $\hat{\theta}$ (see Mean Value Theorem). Solving for $(\hat{\theta}_n - \theta)$ and scaling with $\sqrt{n}$ we have

$$\sqrt{n}(\hat{\theta}_n - \theta) = -\frac{\frac{1}{\sqrt{n}}G_n(\theta)}{\frac{1}{n}G_n'(\tilde{\theta})}$$

which by Slutsky's theorem and CLT gives us

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N_p(0, A^{-1}B(A^T)^{-1})$$

where

$$A = A(\theta) = E\left[\frac{\partial}{\partial\theta^T}G(\theta)\right] \quad \text{and} \quad B = B(\theta) = \text{Var}(G(\theta)).$$

(don't worry about understanding the full derivations right now, Dr. Datta will explain in great detail in LST. It may be helpful to view G as a score function, or something relatively close to it.) In practice we replace A and B by their estimates for inference, that is

$$\hat{A}(\hat{\theta}_n) = \frac{1}{n}\sum_{i=1}^{n}\frac{\partial}{\partial\theta}\Big[G(\theta)\Big]_{\theta=\hat{\theta}_n} \quad \text{and} \quad \hat{B}(\hat{\theta}_n) = \frac{1}{n}\sum_{i=1}^{n}\Big[G(\theta)G(\theta)^T\Big]_{\theta=\hat{\theta}_n}.$$

and the asymptotic distribution can be expressed as

$$[\hat{A}(\hat{\theta}_n)^{-1}\hat{B}(\hat{\theta}_n)(\hat{A}(\hat{\theta}_n)^T)^{-1}]^{-1/2}\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, I_p)$$

where I_p is the p-dimensional identity matrix.

In the situation where we do not have identical distributions the equations change slightly and the asymptotic distribution can be expressed as

$$[A_n(\hat{\theta}_n)^{-1}B_n(\hat{\theta}_n)(A_n(\hat{\theta}_n)^T)^{-1}]^{-1/2}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, I_p)$$

Note that the iid case is a special case where $A_n = nA$ and $B_n = nB$. In this special case it can be seen that the above equation is equal to the previous equation. This course does not cover the non-iid case, so don't fret.

1.3.1 Delta Method

We can use delta method to then find estimations of parameters of different continuous functions $g(\cdot)$, i.e. we can find the form of the parameter estimate we want. Suppose

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N_p(0, V(\theta))$$

the by delta method we have

$$\sqrt{n}(g(\hat{\theta}_n) - g(\theta)) \xrightarrow{d} N(0, g'(\theta)V(\theta)g'(\theta)^T).$$

1.3.2 Necessary assumptions

Estimating equations can be constructed and used in a number of ways. For each differing way, there are a number of assumptions one has to make. For example, if we wanted to estimate a <u>full model</u> we would need to specify the probability distribution to evaluate the expectations required in the calculations of A and B (Ex. maximum likelihood estimation); robustness of inference and model misspecification need to be examined.

If we wanted to estimate <u>distribution free quantities</u> (e.g. the first two moments) we require minimal assumptions. Expectations required to evaluate variance of $\hat{\theta}$ are calculated empirically from the observed data using either sandwich or bootstrap methods.

1.4 Examples: Statistics and Delta Method

1. For $i = 1, \ldots, n$, let X_i be iid random variables with $E(X_i) = \mu$ and $Var(X_i) = \sigma^2$. Approximate the distribution of $1/\bar{X}$ with $\bar{X} = \frac{1}{n}\sum X_i$.

(1) Delta method, by definition, states that if we have a sufficiently continuous function, $g(\cdot)$, (i.e. it is continuous on the sample space) then we can represent the distribution of a estimated sample statistic, $\hat{X}$, as follows:

$$\sqrt{n}(g(\hat{X}) - g(X)) \xrightarrow{d} N(0, g'(X)V(\hat{X})g'(X)^T)$$

where $V(\hat{X})$ is the variance of $\hat{X}$, and $\hat{X}$ is the estimate of X. In the situation here, we know that given any X_i (we will use X_1 since the X's are iid), $E(X_1) = \mu$, and $\text{Var}(X_1) = \sigma^2$. So, we have that

$$\sqrt{n}(\bar{X} - \mu) \xrightarrow{d} N(0, \sigma^2)$$

If we let $g(x) = \frac{1}{x}$, then all we have to do is calculate $g'(x) = \frac{-1}{x^2}$ and apply the formula. Hence,

$$\sqrt{n}\left(\frac{1}{\bar{X}} - \frac{1}{\mu}\right) \xrightarrow{d} N\left(0, \frac{\sigma^2}{\mu^4}\right).$$

If we wanted the direct asymptotic distribution, we can instead write

$$\frac{1}{\bar{X}} \sim N\left(\frac{1}{\mu}, \frac{\sigma^2}{n\mu^4}\right).$$

2. In a linear regression,

$$(Y_i|X_i = x_i) \sim N(\beta_0 + \beta_1 x_i, \sigma^2), \quad i = 1, \ldots, n$$

Show that the sufficient statistics for $(\beta_0, \beta_1, \sigma^2)$ are $(\hat{\beta}_0, \hat{\beta}_1, \hat{\sigma^2}, \sum X_i, \sum X_i^2)$ with

$$\begin{aligned}
\hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} \\
\hat{\beta}_1 &= \sum (x_i - \bar{x})(y_i - \bar{y}) / \sum (x_i - \bar{x})^2 \\
\hat{\sigma^2} &= \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 / (n-2).
\end{aligned}$$

(2) The situation describes a normal distribution where the mean is represented by a linear regression model. That is,

$$\mu_i = \beta_0 + \beta_1 x_i$$

We can fully write the normal distribution as such,

$$p(y_i; \beta_0, \beta_1, \sigma^2) = (2\pi\sigma^2)^{-1/2} \exp\left(\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{-2\sigma^2}\right)$$

We're going to utilize **Fisher-Neyman Factorization Theorem** here, which states that $T(\mathbf{Y})$ is a sufficient statistic *iff* $p(\mathbf{Y}|\theta) = g[T(\mathbf{Y})|\theta] \times h(\mathbf{Y})$. Therefore, the idea is to just separate the equation into one with the sufficient statistic we wish to identify, and the other with the remaining stuff. We'll use some funky math magic to do this.

$$\begin{aligned}
p(y_i; \beta_0, \beta_1, \sigma^2) &= (2\pi\sigma^2)^{-1/2} \exp\left(\frac{(y_i - (\beta_0 + \beta_1 x_i))^2}{-2\sigma^2}\right) \\
p(\boldsymbol{y_i}; \beta_0, \beta_1, \sigma^2) &= \prod_{i=1}^{n} (2\pi\sigma^2)^{-1/2} \exp\left(\frac{y_i^2 - 2y_i\beta_0 - 2y_i\beta_1 x_i + \beta_0^2 + 2\beta_0\beta_1 x_i + \beta_1^2 x_i^2}{-2\sigma^2}\right) \\
&= (2\pi\sigma^2)^{-n/2}. \\
&\quad \exp\left(\frac{\sum y_i^2 - 2\beta_0 \sum y_i - 2\beta_1 \sum x_i y_i + n\beta_0^2 + 2\beta_0\beta_1 \sum x_i + \beta_1^2 \sum x_i^2}{-2\sigma^2}\right) \\
&= (2\pi\sigma^2)^{-n/2} \cdot \exp\left(\frac{n}{-2\sigma^2}\left(\overline{y^2} - 2\beta_0\overline{y} - 2\beta_1\overline{xy} + \beta_0^2 + 2\beta_0\beta_1\overline{x} + \beta_1^2\overline{x^2}\right)\right)
\end{aligned}$$

Isolating β_0

$$= (2\pi\sigma^2)^{-n/2} \cdot \exp\left(\frac{n}{-2\sigma^2}\left(\overline{y^2} - 2\beta_1\overline{xy} + \beta_1^2\overline{x^2}\right)\right) \times$$
$$\exp\left(\frac{n}{\sigma^2}\left(\frac{1}{2}\beta_0^2 + \beta_0 \underbrace{(\overline{y} - \beta_1\overline{x})}_{\text{the sufficient statistic}}\right)\right)$$

Isolating β_1

$$\begin{aligned}
&= (2\pi\sigma^2)^{-n/2} \cdot \exp\left(\frac{n}{-2\sigma^2}\left(\overline{y^2} - 2\beta_0\overline{y} + \beta_0^2\right)\right) \times \\
&\quad \exp\left(\frac{n}{-2\sigma^2}\left(\beta_1^2\overline{x^2} - 2\beta_1\overline{xy} + 2\beta_0\beta_1\overline{x}\right)\right) \\
&= (2\pi\sigma^2)^{-n/2} \cdot \exp\left(\frac{n}{-2\sigma^2}\left(\overline{y^2} - 2\beta_0\overline{y} + \beta_0^2\right)\right) \times \\
&\quad \exp\left(\frac{\beta_1}{-2\sigma^2}\left(\beta_1\sum(x_i^2) - 2\sum(x_iy_i) + 2\sum x_i\sum y_i - \beta_1\left(\sum x_i\right)^2\right)\right) \\
&= (2\pi\sigma^2)^{-n/2} \cdot \exp\left(\frac{n}{-2\sigma^2}\left(\overline{y^2} - 2\beta_0\overline{y} + \beta_0^2\right)\right) \times \\
&\quad \exp\left(\frac{\beta_1}{-2\sigma^2}\left(\beta_1\sum(x_i - \overline{x})^2 - 2\sum(x_i - \overline{x})(y_i - \overline{y})\right)\right) \\
&= (2\pi\sigma^2)^{-n/2} \cdot \exp\left(\frac{n}{-2\sigma^2}\left(\overline{y^2} - 2\beta_0\overline{y} + \beta_0^2\right)\right) \times \\
&\quad \exp\left(\frac{\beta_1}{\sigma^2}\underbrace{\left(\sum(x_i^2) - \overline{x}^2\right)}_{\text{suff. stats}}\left(\frac{-1}{2}\beta_1 + \underbrace{\frac{\sum(x_i - \overline{x})(y_i - \overline{y})}{\sum(x_i - \overline{x})^2}}_{\text{suff. stats}}\right)\right)
\end{aligned}$$

Isolating σ^2

$$\begin{aligned}
&= (2\pi\sigma^2)^{-n/2}\exp\left(\sum\frac{(y_i - (\beta_0 + \beta_1x_i))^2}{-2\sigma^2}\right) \\
&= (2\pi\sigma^2)^{-n/2}\exp\left(\frac{(n-2)}{-2\sigma^2}\underbrace{\sum\frac{(y_i - (\beta_0 + \beta_1x_i))^2}{n-2}}_{\text{suff. stats}}\right)
\end{aligned}$$

These estimations can also be found by using maximum likelihood estimations. In the course a major proposition is ignored and not explained or proved. The proposition states that, under regularity conditions, maximum likelihood estimators contain all sufficient statistics for their respective parameter. See this link for a proof & explanation. In the next section we'll use MLE estimators to show that these are the correct sufficient statistics.

1.5 Likelihood

It's important to understand what likelihood is the measure of to begin with. In Cramer's book "Mathematical Methods of Statistics" (It's very dry, although I do recommend it) he explains that by combining the equations of probability for n many random variables, we arrive at the likelihood function whose input is now the parameter estimate, and whose defined space is based on the random variables. Let me use the Poisson distribution as an example. Consider $Y = \{y_1, \ldots, y_n\}$ such that $y_i \sim Poisson(\lambda)$ for all $i \in 1 : n$. Then we have,

$$p(y_i; \lambda) = exp(-\lambda + y \log(\lambda) - C_i)$$

where $C_i = \log(y_i!)$. This is clearly a function of y_i over a space defined by the distribution. That is, the domain of the function is $y_i \in [0, \infty)$ and it depends on the choice of parameter λ. However, if we analyze the joint probability of all y_i's,

$$p(\mathbf{y}; \lambda) = exp(-n\lambda + n\overline{y} \log(\lambda) - n\overline{C})$$

we see that the domain of the function now always contains all possible random variables y_i for the n samples. It doesn't make sense to use values of y_i as a domain anymore, because we are spanning the entire domain of y at once. However, the parameter λ may take many different values over the random variables. If we swap the input and parameter such that now we are measuring the value of the function over λ, we can find the *likelihood* that said parameter value is correct for the provided sample of random variables. It follows that the highest value that the likelihood function takes is the most likely value of the parameter.

$$L(\lambda; \mathbf{y}) = exp(-n\lambda + n\overline{y} \log(\lambda) - n\overline{C})$$

This leads us to the method of maximum likelihood, which was first published by Fisher. If you're still confused, even slightly, watch *this video.*

1.5.1 Maximum Likelihood Estimation

The value of θ that maximizes $L(\theta)$, and therefore gives the highest probability (density) to the observed data, denoted $\hat{\theta}$, is known as the maximum likelihood estimator (MLE).

The goal of maximization allows for a monotonic transformation of the function (a transformation that maintains order of values). It is computationally convenient to consider the log scale of the likelihood function, the *log-likelihood* function.

$$l(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log p(Y_i|\theta)$$

Just as we find maximums in calculus, we use the derivative of the log-likelihood to evaluate the roots. In other words, we can find $\hat{\theta}$ by taking the partial derivative of $l(\theta)$ with respect to θ, setting it equal to zero, then solving for θ. The derivative of the log-likelihood is called the score function:

$$S(\theta) = \frac{\partial l(\theta)}{\partial \theta} = \left[\frac{\partial l(\theta)}{\partial \theta_1}, \ldots, \frac{\partial l(\theta)}{\partial \theta_p}\right]^T = [S(\theta_1), \ldots, S(\theta_p)]$$

which is the $p \times 1$ vector of derivatives of the log-likelihood. It is also in our interest to know whether the function is reaching a maximum or a minimum (roots do not identify this); thus

we take the second derivative. We then take the expected value so that we can determine if the function reaches a maximum or minimum over unlimited samples. The Fisher's expected information for a sample size of n is the $p \times p$ matrix

$$\boldsymbol{I}_n(\theta) = -E\left[\frac{\partial^2}{\partial\theta\partial\theta^T} l(\theta)\right] = -E\left[\frac{\partial S(\theta)}{\partial\theta^T}\right].$$

Under suitable regular conditions,

$$E[S(\theta)] = 0 \quad \text{and} \quad \boldsymbol{I}_n(\theta) = E[S(\theta)S(\theta)^T]$$

proofs of the above statements are provided in the textbook and can be found easily with the definition of expectation for continuous distributions.
Regular conditions:

1. Parameters in Θ (the parameter space) are identifiable.
2. The range of data is independent of the parameters.
3. Parameter values lie in the interior of the parameter space.

The score function can be viewed as an estimating functions and is applied in the exact same way as described above:

$$G_n(\theta) = \frac{1}{n} S(\theta),$$

which under regular conditions satisfies our conditions that

$$E[G_n(\theta)] = 0.$$

1.5.2 Variants on Likelihood:

1. Conditional Likelihood
   (a) Only applies to a particular situation
   (b) A probability and has similar properties to regular likelihood
2. Marginal Likelihood
   (a) Only applies to a particular situation
   (b) A probability and has similar properties to regular likelihood
3. Profile Likelihood
   (a) Always applies
   (b) Not a probability

All variants provide a strategy to hand the nuisance parameters.

1.5.3 Conditional Likelihood

Suppose λ represent parameters of interest, with ϕ being nuisance parameters. Suppose the distribution for y can be factorized as

$$p(\mathbf{y}|\lambda, \phi) \propto p(t_1|t_2, \lambda)p(t_2|\lambda, \phi),$$

where t_1 and t_2 are (not ancillary) statistics, that is, functions of $\mathbf{y}$. Then inference for λ may be based on the *conditional likelihood*

$$L_c(\lambda) = p(t_1|t_2, \lambda).$$

1.5.4 Marginal Likelihood

Let S_1, S_2, A be a minimal sufficient statistic where A is ancillary and suppose we have the factorization

$$p(y|\lambda, \phi) \propto p(s_1, s_2, a|\lambda, \phi) = p(a)p(s_1|a, \lambda)p(s_2|s_1, a, \lambda, \phi).$$

Inference for λ may be based on the *marginal likelihood*

$$L_m(\lambda) = p(s_1|a, \lambda).$$

1.5.5 Profile Likelihood

Profile likelihood provides a method of examining the behavior of a subset of the parameters. If $\theta = [\lambda, \phi]$, where λ again represents a vector of parameters of interest and ϕ the remaining parameters, then the profile likelihood $L_p(\lambda)$ for λ is defined as

$$L_p(\lambda) = \max_{\phi} L(\lambda, \phi).$$

If $\tilde{\lambda}$ denotes the maximum of $L_p(\lambda)$ and $\hat{\theta} = [\hat{\lambda}, \hat{\phi}]$ is the MLE, then $\tilde{\lambda} = \hat{\lambda}$. While we can always apply this technique, we cannot treat it like other likelihoods (as a probability); however it is useful for variance component estimation in mixed effect models.

1.6 Quasi-Likelihood

1.6.1 Maximum Quasi-Likelihood Estimation (MQLE)

Quasi likelihood assumes that we have differing variance per random variable; they may be identically distributed, but not independent. There may be some conditional variation between the samples causing them to have differing variances.

For quasi likelihood we only need the first two moments. The mean model is readily available in a regression setting. The variance model assumed a mean-variance relationship. The estimating function used to describe quasi likelihood is based upon the mean and variance of the data only. Specifically, we assume that the first two moments are of the form

$$\begin{aligned} E[Y_i|\beta] &= \mu_i(\beta) \\ \text{Var}(Y_i|\beta) &= \alpha V[\mu_i(\beta)] \end{aligned}$$

where $\mu_i(\beta) = X_i^T\beta$ is the mean response given a single set of the predictors, of X $(n \times k + 1)$, and the regression coefficients $\beta = \{\beta_0, \ldots, \beta_k\}$, and $V[\mu_i(\beta)]$ is the variance of $\mu_i(\beta)$. It follows that

$$\begin{aligned} E[\boldsymbol{Y}|\beta] &= \boldsymbol{\mu}(\beta) \\ \text{Var}(\boldsymbol{Y}|\beta) &= \alpha \boldsymbol{V}[\boldsymbol{\mu}(\beta)] \end{aligned}$$

as the matrix form, where $\boldsymbol{\mu} = \{\mu_1(\beta), \ldots, \mu_n(\beta)\}$ represents the regression function for $X^T\beta$, and $\boldsymbol{V}[\boldsymbol{\mu}(\beta)]$ as the covariance matrix given the predictors, and $\alpha > 0$ is a scalar that does not depend upon β. We assume $\beta = [\beta_0, \ldots, \beta_k]^T$ so that the dimension of β is $k+1$. To motivate an estimating function, consider the sum of squares

$$(\boldsymbol{Y} - \boldsymbol{\mu})^T \boldsymbol{V}^{-1} (\boldsymbol{Y} - \boldsymbol{\mu})/\alpha$$

where $\boldsymbol{\mu} = \boldsymbol{\mu}(\beta)$ and $\boldsymbol{V} = \boldsymbol{V}(\beta)$. In order to minimize this sum of squares there are two ways to proceed. The first and more obvious route is to differentiate by β, set to zero, and solve to find the minimum.

$$-2\boldsymbol{D}^T\boldsymbol{V}^{-1}(\boldsymbol{Y} - \boldsymbol{\mu})/\alpha + (\boldsymbol{Y} - \boldsymbol{\mu})^T \frac{\partial \boldsymbol{V}^{-1}}{\partial \beta}(\boldsymbol{Y} - \boldsymbol{\mu})/\alpha,$$

where $\boldsymbol{D}$ is the $n \times p$ matrix of derivatives with elements $\partial\mu_i/\partial\beta_j,\ i = 1, \ldots, n\ j = 1, \ldots, p$. However, this has the issue that the resulting derivative does not necessarily have expectation zero when we only assume $E[\boldsymbol{Y}|\beta] = \boldsymbol{\mu}$, because of the presence of the second term. If the expectation of the estimating function is not zero, then an inconsistent estimator of β results. As shorthand we write this estimating function as

$$\boldsymbol{U}(\beta) = \boldsymbol{D}^T\boldsymbol{V}^{-1}(\boldsymbol{Y} - \boldsymbol{\mu})/\alpha + \underbrace{(\boldsymbol{Y} - \boldsymbol{\mu})^T \frac{\partial \boldsymbol{V}^{-1}}{\partial \beta}(\boldsymbol{Y} - \boldsymbol{\mu})/\alpha}_{\text{completely ignored because expectation isn't 0}} .$$

So we have

$$\boldsymbol{U}(\beta) = \boldsymbol{D}^T\boldsymbol{V}^{-1}(\boldsymbol{Y} - \boldsymbol{\mu})/\alpha.$$

This estimating function is linear in the data and so its properties are straightforward to evaluate. In particular,

1. $E[\boldsymbol{U}(\beta)] = 0$, assuming $E[Y|\beta] = \boldsymbol{\mu}(\beta)$.
2. $\text{Var}[\boldsymbol{U}(\beta)] = \boldsymbol{D}^T\boldsymbol{V}^{-1}\boldsymbol{D}/\alpha$, assuming $\text{Var}(\boldsymbol{Y}|\beta) = \alpha V$.
3. $-E[\frac{\partial \boldsymbol{U}}{\partial \beta}] = \boldsymbol{D}^T\boldsymbol{V}^{-1}\boldsymbol{D}/\alpha = \text{Var}[\boldsymbol{U}(\beta)]$, assuming $E[\boldsymbol{Y}|\beta] = \boldsymbol{\mu}(\beta)$.

We can directly apply estimating function equations to get the asymptotic distribution,

$$(\boldsymbol{D}^T\boldsymbol{V}^{-1}\boldsymbol{D})^{1/2}(\hat{\beta}_n - \beta) \xrightarrow{d} N_{k+1}(0, \alpha\mathbf{I}_{k+1})$$

where α is known. If μ were known instead we can estimate $\hat{\alpha}_n$ with,

$$\hat{\alpha}_n = (\boldsymbol{Y} - \boldsymbol{\mu})^T\boldsymbol{V}^{-1}(\boldsymbol{\mu})(\boldsymbol{Y} - \mu)/n.$$

(note: for model based estimation we "know" $\boldsymbol{\mu}$). The degree of freedom corrected version of this is given by,

$$\hat{\alpha}_n = \frac{1}{n-k-1}\sum_{i=1}^{n}\frac{(Y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}.$$

where $\hat{\mu}_i = \hat{\mu}_i(\hat{\beta})$. In practice we have to estimate all statistics and we cannot assume parameter are known. In practice we use the following equation:

$$(\hat{\boldsymbol{D}}^T \hat{\boldsymbol{V}}^{-1} \hat{\boldsymbol{D}}/\hat{\alpha}_n)^{1/2}(\hat{\beta}_n - \beta) \xrightarrow{d} N_{k+1}(0, \mathbf{I}_{k+1}).$$

The models above work regardless of how "correct" the estimations of the mean model are. We simply estimate a specified form of association regardless of model appropriateness. The usefulness of inference doesn't depend on the appropriate model.

1.6.2 Quasi-log-likelihood

We can find the quasi-log-likelihoods of these models and relate them to known distributions using the following equations. If we ignore regression and just consider a function of μ (one dimensional) we will have that $D = \partial\mu/\partial\mu = 1$, which gives us a quasi-score of

$$S_q = \frac{y - \mu}{\alpha V(\mu)}.$$

To really drive this home, let us consider what we're doing. We are taking the sum of squares, taking the derivative, and setting it equal to zero. The reduced function is then the quasi-score. Integration of this quantity over the domain (which is now the sample) up to the value of the function (which is the statistic we want to estimate) gives

$$l(\mu, \alpha) = \int_y^\mu \frac{y - t}{\alpha V(t)} dt$$

which, if it exists, behaves like a log-likelihood. As an example, for the model $E[Y] = \mu$ and $\text{Var}(Y) = \alpha\mu$ we have that

$$\begin{aligned}
l(\mu, \alpha) &= \int_y^\mu \frac{y - t}{\alpha t} dt \\
&= \int_y^\mu \left[\frac{y}{\alpha t} - \frac{1}{\alpha}\right] dt \\
&= \left[\frac{y}{\alpha}\log(t) - \frac{t}{\alpha}\right]_{t=\mu} - \left[\frac{y}{\alpha}\log(t) - \frac{t}{\alpha}\right]_{t=y} \\
&= \frac{1}{\alpha}[y\log(\mu) - \mu - (y\log(y) - y)] \\
&= \frac{1}{\alpha}[y\log(\mu) - \mu - c]
\end{aligned}$$

where $c = (y\log(y) - y)$ is a known constant. Note that $(y\log(\mu) - \mu)$ is the log-likelihood of a Poisson random variable.

The derivation of the above is the derivation of the quasi-likelihood. Let's run through some more examples to clear the confusion, because just like you, when I read this my brain turns off.

Example Let's assume we are working with some regression data for a response variable Y, and we find the relationship that the mean is $E(Y|\beta) = \mu$ and the variance act as a constant, as

if $\text{Var}(Y|\beta) = \alpha$. Remember, the goal here is to **estimate the vector** β. The definition tells us to represent this as

$$\underbrace{E(Y|\beta) = \mu(\beta)}_{\text{definition}} = \underbrace{\mu}_{\text{example}}$$
$$\underbrace{\text{Var}(Y|\beta) = \alpha V(\mu(\beta))}_{\text{definition}} = \underbrace{\alpha \times I_n}_{\text{example}}$$

The sum of squares in this case can be represented as

$$\underbrace{(Y-\mu)^T V^{-1}(Y-\mu)/\alpha}_{\text{definition}} = \underbrace{(Y-\mu)^T(Y-\mu)/\alpha}_{\text{example}}$$

To find the minimum of the sum of squares we want to take the derivative and set to zero. This gives us the estimating function (think score):

$$U(\beta) = \underbrace{-2D^T V^{-1}(Y-\mu)/\alpha}_{\text{definition}} = \underbrace{-2D^T(Y-\mu)/\alpha}_{\text{example}}$$

where D is the $n \times p$ matrix of derivatives with elements $\partial\mu_i/\partial\beta_j,\ i = 1, \ldots, n\ j = 1, \ldots, p$. So now we have an estimating function. We treat estimating functions in the same way we described them in section 1.3. So, to estimate β we just follow the definition of estimating functions, that is:

$$\underbrace{(D^T V^{-1} D)^{1/2}(\hat{\beta}_n - \beta) \xrightarrow{d} N_{k+1}(0, \alpha \mathbf{I}_{k+1})}_{\text{definition}}$$
$$\underbrace{(D^T D)^{1/2}(\hat{\beta}_n - \beta) \xrightarrow{d} N_{k+1}(0, \alpha \mathbf{I}_{k+1})}_{\text{example}}$$

It's that easy. Just follow the definitions. In practice the solution would follow the estimated variance and estimated α. Hence we would have

$$(\hat{D}^T \hat{V}^{-1} \hat{D}/\hat{\alpha}_n)^{1/2}(\hat{\beta}_n - \beta) \xrightarrow{d} N_{k+1}(0, \mathbf{I}_{k+1}).$$

which, once again, is just definition application.

1.6.3 Maximum Quasi-log-likelihood estimations for common distributions

Now, let's ignore the regression and relate the quasi likelihood directly to distributions. Let's run through the examples from the book and correct them. Each of these examples takes $E(Y) = \mu$.

Example 1. Show that $Var(Y) = 1 \implies N(\mu, \alpha)$

$$
\begin{aligned}
S_q &= \frac{y-\mu}{\alpha} \\
l(\mu, \alpha) &= \int_y^\mu \frac{y-t}{\alpha} dt \\
&= \left[\frac{yt}{\alpha} - \frac{t^2}{2\alpha}\right]_{t=\mu} - \left[\frac{yt}{\alpha} - \frac{t^2}{2\alpha}\right]_{t=y} \\
&= \frac{y\mu}{\alpha} - \frac{\mu^2}{2\alpha} - \frac{y^2}{\alpha} + \frac{y^2}{2\alpha} \\
&= -\frac{1}{2\alpha}\left[(\mu^2 - 2y\mu + y^2)\right] \\
&= -\frac{1}{2\alpha}\left[(y-\mu)^2\right]
\end{aligned}
$$

This resulting likelihood corresponds to the log-likelihood we would get if we derived the log-likelihood for the distribution $N(\mu, \alpha)$. This can be seen when we compare the score to the quasi-score. See,

$$
\begin{aligned}
p(\mathbf{y}) &= (2\pi\alpha)^{-n/2} \exp(-\frac{1}{2\alpha}\sum(y-\mu)^2) \\
l(\mu, \alpha) &= (-n/2)\log(2\pi\alpha) - \frac{1}{2\alpha}\sum(y-\mu)^2
\end{aligned}
$$

for univariate y

$$
S_1(\mu) = \frac{y-\mu}{\alpha} \equiv S_q.
$$

Example 2. Show that $Var(Y) = \mu^2 \implies Ga(1/\alpha, 1/\mu\alpha)$

$$
\begin{aligned}
S_q &= \frac{y-\mu}{\alpha\mu^2} \\
l(\mu, \alpha) &= \frac{1}{\alpha}\int_y^\mu \left[\frac{y}{t^2} - \frac{1}{t}\right] dt \\
&= \frac{1}{\alpha}\left[-\frac{y}{t} - \log(t)\right]_{t=\mu} - \frac{1}{\alpha}\left[-\frac{y}{t} - \log(t)\right]_{t=y} \\
&= \frac{1}{\alpha}\left[-\frac{y}{\mu} - \log(\mu)\right] - \frac{1}{\alpha}\left[-\frac{y}{y} - \log(y)\right] \\
&= \frac{1}{\alpha}\left[-\frac{y}{\mu} - \log(\mu) + 1 + \log(y)\right] \\
&= \frac{1}{\alpha}\left[\frac{-y}{\mu} - \log(\mu)\right] + [\text{constants}]
\end{aligned}
$$

For $y \sim Ga(1/\alpha, 1/\mu\alpha)$,

$$p(\mathbf{y}) = \left[\frac{(1/\mu\alpha)^{(1/\alpha)}}{\Gamma(1/\alpha)}\right]^n \prod \left[y_i^{(1/\alpha)-1}\right] \exp\left(-\left(\sum y_i\right)/\mu\alpha\right)$$
$$l = n \log\left[\frac{(1/\mu\alpha)^{(1/\alpha)}}{\Gamma(1/\alpha)}\right] + [1/\alpha - 1]\sum \log(y_i) - \left(\sum y_i\right)/\mu\alpha$$

for univariate y

$$S_1(\mu) = \frac{y}{\mu^2\alpha} - \frac{1}{\mu\alpha} = \frac{y-\mu}{\alpha\mu^2} \equiv S_q$$

to get the derivative in the first term of the log-likelihood, use either wolfram or u-sub.

The other distributions in the table on page 52 can be done on your own.

1.7 Sandwich Estimation

A general method of avoiding stringent modeling conditions when the variance of an estimator is calculated is provided by *sandwich estimation.* Consider the estimating equations discussed in section 1.3. We have

$$G_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} G(\theta, Y_i)$$

with

$$A = E\left[\frac{\partial}{\partial\theta^T} G(\theta, Y)\right] \quad \text{and} \quad B = E\left[G(\theta, Y)G(\theta, Y)^T\right].$$

Based on iid observations we have the sandwich form of the variance as

$$\text{Var}(\hat{\theta}_n) = \frac{A^{-1}B(A^T)^{-1}}{n}.$$

We can estimate A, B, and subsequently $\text{Var}(\hat{\theta}_n)$ emperically, as we've done before.

$$\hat{A}_n = \frac{1}{n}\sum_{i=1}^{n} \frac{\partial}{\partial\theta} G(\hat{\theta}, Y_i)$$
$$\hat{B}_n = \frac{1}{n}\sum_{i=1}^{n} G(\hat{\theta}, Y_i)G(\hat{\theta}, Y_i)^T$$
$$\hat{\text{Var}}(\hat{\theta}_n) = \frac{\hat{A}_n^{-1}\hat{B}_n(\hat{A}_n^T)^{-1}}{n}$$

By weak law of large numbers $\hat{A}_n \xrightarrow{p} A$ and $\hat{B}_n \xrightarrow{p} B$ so $\hat{\text{Var}}(\hat{\theta}_n)$ is a consistent estimator of the variance.

Now if we consider the same iid situation where the score is the estimating equation. Let l_i be

the log-likelihood of random variable Y. We get

$$\begin{aligned}
G_n(\theta) =& \frac{1}{n}\sum_{i=1}^{n}\frac{\partial}{\partial\theta}l_i(\theta) \\
A =& \frac{1}{n}\sum_{i=1}^{n}E\left[\frac{\partial^2}{\partial\theta\partial\theta^T}l(\theta)\right] \\
B =& \frac{1}{n}\sum_{i=1}^{n}E\left[\left(\frac{\partial}{\partial\theta}l(\theta)\right)\left(\frac{\partial}{\partial\theta}l(\theta)\right)^T\right].
\end{aligned}$$

We can estimate the sandwich estimator in this fashion by setting

$$\begin{aligned}
\hat{A}_n =& \frac{1}{n}\sum_{i=1}^{n}\frac{\partial^2}{\partial\theta\partial\theta^T}l(\theta)\bigg|_{\theta=\hat{\theta}} \\
\hat{B}_n =& \frac{1}{n}\sum_{i=1}^{n}\left(\frac{\partial}{\partial\theta}l(\theta)\right)\left(\frac{\partial}{\partial\theta}l(\theta)\right)^T\bigg|_{\theta=\hat{\theta}}
\end{aligned}$$

which gives the following **estimator for variance**

$$\hat{\mathrm{Var}}(\hat{\theta}_n) = \frac{\hat{A}_n^{-1}\hat{B}_n(\hat{A}_n^T)^{-1}}{n}$$

We can apply this method to the estimating functions we found in the previous section about quasi-likelihood. The estimating function we had previously found was

$$U(\beta) = D^TV^{-1}(Y-\mu)/\alpha.$$

By the same design,

$$\begin{aligned}
G_n(\theta) &= U(\beta) \\
A &= E\left[\frac{\partial}{\partial\beta^T}U(\beta)\right] \\
B &= E\left[U(\beta)U(\beta)^T\right]
\end{aligned}$$

and the sandwich estimator and it's estimation follow.

1.8 Hypothesis Testing

Motivation The motivation of hypothesis testing coincides with the motivation of estimation. We want to know if the estimate of a statistic is good or bad. A test can provide inference to this result. For example, in a regression setting, if the interval in which a parameter is estimated contains 0 with 95% confidence, then the resulting test of if the estimator is in fact null (zero) will show the same result, that the parameter is likely to not matter.

Partitioned Testing In some environments we have multivariate θ and wish to test $r < p$ many parameters rather than all p. We may then take the parameters and order them such that $\theta = \{\theta_1, \ldots, \theta_{p-r}, \theta_{p-r+1}, \ldots, \theta_p\}$. Then, the information is formed such that (for the independent case)

$$I_1 = \begin{pmatrix} I_{11} & I_{12} \\ I_{21} & I_{22} \end{pmatrix}$$

where $I_{11}, I_{12}, I_{21}, I_{22}$ are of dimensions $(p-r) \times (p-r)$, $(p-r) \times r$, $r \times (p-r)$, and $r \times r$ respectively. The inverse of I_1 is,

$$I_1^{-1} = \begin{pmatrix} I_{11\cdot 2}^{-}1 & -I_{11\cdot 2}^{-1} I_{12} I_{22}^{-1} \\ -I_{22\cdot 1}^{-1} I_{21} I_{11}^{-1} & I_{22\cdot 1}^{-1} \end{pmatrix}$$

where

$$I_{11\cdot 2} = I_{11} - I_{12} I_{22}^{-1} I_{21}$$
$$I_{22\cdot 1} = I_{22} - I_{21} I_{11}^{-1} I_{12}$$

A hypothesis related to this would be

$$H_0 : \begin{cases} \hat{\theta}_1 \text{unrestricted} \\ \hat{\theta}_2 = \theta_2 \end{cases} \qquad H_1 : \hat{\theta} = [\hat{\theta}_1, \hat{\theta}_2] \neq [\hat{\theta}_1, \theta_2]$$

where $\theta_1 = \{\theta_1, \ldots, \theta_{p-r}\}$ and $\theta_2 = \{\theta_{p-r+1}, \ldots, \theta_p\}$.

1.8.1 Initial Derivation: Taylor's Expansion

Both the Score test and Wald test, and their relation to the Likelihood Ratio test, stem from the Taylor expansion of the estimating equation (score in this case).

Let's consider univariate θ. Assume we have a vector of iid random variables, $\mathbf{Y}$, with joint density $f(\mathbf{Y}; \theta)$ with parameter θ and estimator $\hat{\theta}$ such that $\hat{\theta}$ is consistent (i.e. $E(\hat{\theta}) \to \theta$). Define $S(\hat{\theta})$ as the score function where

$$S_n(\hat{\theta}) = \sum_{i=1}^{n} \frac{\partial}{\partial \theta} l(\theta) \bigg|_{\theta = \hat{\theta}}$$

where $l(\theta)$ is the corresponding log-likelihood function for θ. Assume also that

$$E(S_n(\hat{\theta})) = \int \left[\sum_{i=1}^{n} \frac{1}{f(y_i; \theta)} \frac{\partial}{\partial \theta} f(y_i; \theta) \right] f(y; \theta) dy = \sum_{i=1}^{n} \frac{\partial}{\partial \theta} \int f(y_i; \theta) dy = 0$$

and

$$-E\left(\frac{\partial}{\partial \theta} S_n(\theta) \right) = I_n(\theta) \underbrace{=}_{\text{iid case}} n I_1(\theta) \underbrace{=}_{\text{single sample}} I_1(\theta).$$

The Taylor expansion of $S_n(\hat{\theta})$ around θ is,

$$0 = S_n(\hat{\theta}) = S_n(\theta) + (\hat{\theta} - \theta) \frac{\partial S_n(\tilde{\theta})}{\partial \theta}$$

where $\tilde{\theta}$ is between θ and $\hat{\theta}$ and we require that $\tilde{\theta} \xrightarrow{u} \theta$. With regular conditions, $\frac{\partial S_n(\tilde{\theta})}{\partial \theta} \to I_n(\theta)$, so the notation will be relaxed. This next equation is the basis of each test.

$$\frac{\frac{1}{\sqrt{n}} S_n(\theta)}{I_1(\theta)} \approx \sqrt{n}(\hat{\theta} - \theta)$$

1.8.2 Score Test

In the score test we solve for the asymptotic distribution of the score function. We derive that

$$\frac{\frac{1}{\sqrt{n}}S_n(\theta)}{I_1(\theta)} \approx \sqrt{n}(\hat{\theta}-\theta) \xrightarrow{d} N(0, I_1^{-1}(\theta)).$$

To estimate, replace θ with $\hat{\theta}$ on the left-hand side. Hence we have that,

$$\frac{\frac{1}{\sqrt{n}}S_n(\theta)}{I_1(\theta)} \xrightarrow{d} N(0, I_1^{-1}(\theta)) \implies \frac{\frac{1}{\sqrt{n}}S_n(\theta)}{\sqrt{I_1(\theta)}} \xrightarrow{d} N(0,1).$$

If we square the L.H.S. we have the final form,

$$S_n(\theta)^T I_1^{-1}(\theta) S_n(\theta)/n \xrightarrow{d} \chi_1^2.$$

For multivariate $\theta = \{\theta_1, \ldots, \theta_p\}$ it is the case that,

$$S_n(\theta)^T I_1^{-1}(\theta) S_n(\theta)/n \xrightarrow{d} \chi_p^2.$$

In the partitioned testing case with $\theta_1 = \{\theta_1, \ldots, \theta_{p-r}\}$ and $\theta_2 = \{\theta_{p-r+1}, \ldots, \theta_p\}$, and hypotheses

$$H_0 : \begin{cases} \hat{\theta}_1 \text{unrestricted} \\ \hat{\theta}_2 = 0 \end{cases} \qquad H_1 : \hat{\theta} = [\hat{\theta}_1, \hat{\theta}_2] \neq [\hat{\theta}_1, 0]$$

we have that,

$$S_n(\theta)^T I_{11\cdot 2}^{-1}(\theta) S_n(\theta)/n \xrightarrow{d} \chi_r^2.$$

1.8.3 Wald Test

The Wald test solves for the asymptotic distribution of the score fraction. To estimate, replace θ with $\hat{\theta}$ on the left-hand side.

$$\sqrt{n}(\hat{\theta}-\theta) \approx \frac{\frac{1}{\sqrt{n}}S_n(\theta)}{I_1(\theta)} \xrightarrow{d} N(0, I_1^{-1}(\theta)).$$

which gives,

$$\sqrt{n} I_1^{(1/2)}(\theta)(\hat{\theta}-\theta) \xrightarrow{d} N(0,1).$$

If we square the L.H.S. we have the final form,

$$n(\hat{\theta}-\theta)^T I_1(\theta)(\hat{\theta}-\theta) \xrightarrow{d} \chi_1^2.$$

For multivariate $\theta = \{\theta_1, \ldots, \theta_p\}$ it is the case that,

$$n(\hat{\theta}-\theta)^T I_1(\theta)(\hat{\theta}-\theta) \xrightarrow{d} \chi_p^2.$$

In the partitioned testing case with $\theta_1 = \{\theta_1, \ldots, \theta_{p-r}\}$ and $\theta_2 = \{\theta_{p-r+1}, \ldots, \theta_p\}$, and hypotheses

$$H_0 : \begin{cases} \hat{\theta}_1 \text{unrestricted} \\ \hat{\theta}_2 = 0 \end{cases} \qquad H_1 : \hat{\theta} = [\hat{\theta}_1, \hat{\theta}_2] \neq [\hat{\theta}_1, 0]$$

we have that,

$$n(\hat{\theta}-\theta)^T I_{11\cdot 2}(\theta)(\hat{\theta}-\theta) \xrightarrow{d} \chi_r^2.$$

1.8.4 Likelihood Ratio Test

The likelihood ratio test is a slight deviation from the Taylor expansion of the score function. Instead of the Taylor expansion of the score function, we instead look at the Taylor expansion of the log-likelihood function directly.

$$l_n(\theta) = l_n(\hat{\theta}) + \underbrace{(\theta - \hat{\theta})\left[\frac{\partial}{\partial\theta} l_n(\theta)\right]_{\hat{\theta}=\theta}}_{\text{becomes 0}} + \underbrace{\frac{1}{2}(\theta - \hat{\theta})^T \left[\frac{\partial^2}{\partial\theta\partial\theta^T} l_n(\theta)\right]_{\hat{\theta}=\tilde{\theta}} (\theta - \hat{\theta})}_{\text{asymptotically equivalent to the final form of the Wald test.}}$$

$$2[l_n(\theta) - l_n(\hat{\theta})] \approx n(\theta - \hat{\theta})^T I_1(\theta)(\theta - \hat{\theta}) \xrightarrow{d} \chi_1^2$$

where $\tilde{\theta}$ is between θ and $\hat{\theta}$ and we require that $\tilde{\theta} \xrightarrow{u} \theta$. Hence we derive the following for the multivariate case,

$$-2[l_n(\hat{\theta}) - l_n(\theta)] \xrightarrow{d} \chi_p^2$$

Quasi-Likelihood Ratio Test It can be noted that the quasi-likelihood ratio test can follow directly from the use of the log-quasi-likelihood as a replacement for log-likelihood in the ratio test formula.

1.8.5 Re-parameterization

It is the case that the score and likelihood ratio tests are both invariant to re-parameterization. The Wald test, however, is not.

2 Bayesian Inference

"In the Bayesian approach to inference, all *unknown* quantities contained in a probability model for the observed data are treated as random variables." (Wakefield)

2.1 Likelihood, Posterior, and Prior

Let $\theta = \{\theta_1, \ldots, \theta_p\}^T$ denote all unknowns of a model, and $y = \{y_1, \ldots, y_n\}^T$ be the observed data vector. Also, let $\mathcal{I}$ be all relevant information that is currently available to the individual. Bayesian inference is based on the *posterior* probability distribution of θ after observing y, which is given by Bayes theorem:

$$p(\theta|y, \mathcal{I}) = \frac{p(y|\theta, \mathcal{I})\pi(\theta|\mathcal{I})}{p(y|\mathcal{I})},$$

where $p(y|\theta, \mathcal{I})$ is the *likelihood* function of y, and $\pi(\theta|\mathcal{I})$ is the *prior* of θ. The prior represents the information we knew about θ before observing the current data (for instance, if θ is a variance parameter that we believe came from a Gaussian distribution, we may set $\theta = s^2$). The term $p(y|\mathcal{I})$ is simply a remnant of the Bayes equation you're probably familiar with; ($P(A|B)P(B) = P(B|A)P(A)$). It keeps the likelihood and prior within the required parameter space. We will suppress the dependence on I,

$$p(\theta|y) = \frac{p(y|\theta)\pi(\theta)}{p(y)},$$

where the normalizing constant is

$$p(y) = \int_\theta p(y|\theta)\pi(\theta)d\theta$$

and is the marginal probability of the observed data given the model. Ignoring the constant gives,

$$p(\theta|y, \mathcal{I}) \propto p(y|\theta, \mathcal{I})\pi(\theta|\mathcal{I}) \equiv \textbf{Posterior} \propto \textbf{Likelihood} \times \textbf{Prior}.$$

Because we depend on updating the prior after every iteration of data retrieval, it's important to see how sequential data forms within the equations. Let y_1 and y_2 be the first and second data sets (in order), respectively. We want to know what the full equation

$$p(\theta|y_1, y_2) = \frac{p(y_1, y_2|\theta)\pi(\theta)}{p(y_1, y_2)},$$

will become. Consider the time when only the data y_1 was available.

$$p(\theta|y_1) = \frac{p(y_1|\theta)\pi(\theta)}{p(y_1)}$$

Hence when we observe y_2, the prior depends on the information obtained from y_1.

$$p(\theta|y_1, y_2) = \frac{p(y_2|y_1, \theta)\pi(\theta|y_1)}{p(y_2|y_1)}$$

following Bayes formula, $\pi(\theta|y_1) = p(y_1|\theta)\pi(\theta)/p(y_1)$, we can substitute in the equation and get,

$$p(\theta|y_1, y_2) = \frac{p(y_2|y_1, \theta)}{p(y_2|y_1)} \times \frac{p(y_1|\theta)}{p(y_1)} \times \pi(\theta).$$

If we are in the case that the data collected is conditionally independent, then the formula becomes to,

$$p(\theta|y_1, y_2) = \frac{p(y_2|\theta)}{p(y_2)} \times \frac{p(y_1|\theta)}{p(y_1)} \times \pi(\theta).$$

From both results it can be seen that we reach a consistent inference on y_1 and y_2 regardless of if we choose to use one or two stages to create the prior; its form still appears isolated.

2.1.1 more on the posterior

Typically the posterior distribution is multivariate, and marginal distributions for parameters of interest will be needed. The **univariate marginal distribution** for θ_i is

$$p(\theta_i|y) = \int_{\theta_{-i}} p(\theta|y) d\theta_{-i}$$

where $\theta = \{\theta_1, \ldots, \theta_{i-1}, \theta_{i+1}, \ldots, \theta_p\}$. It must be understood that once we have the marginal distribution of the parameter of interest, we can use that to find the following information about the parameter. The **posterior mean** from this point is simply

$$E[\theta_i|y] = \int_{\theta_i} \theta_i p(\theta_i|y) d\theta_i.$$

Quantiles can be formed from this point as well,

$$q = \Pr[\theta_i \leq \theta_i(q)] = \int_{-\infty}^{\theta_i(q)} p(\theta_i|y) d\theta_i.$$

The posterior mean is quite a useful indicator for location. We can judge the value of a parameter estimate equation by using the expected loss. Let $L(\theta, a)$ denote the loss associated with choosing action a when θ is the true state of nature. The **expected loss** of an action a is then,

$$\overline{L}(a) = \int_{\theta} L(\theta, a) p(\theta|y) d\theta.$$

The optimal choice of action is one that minimized the expected loss equation above. We can use different loss functions to find different optimal estimates. For example, quadratic loss, $L(\theta, a) = (\theta - a)^2$, leads to the posterior mean $\hat{a} = E[\theta|y]$. Linear loss

$$L(\theta, a) = \begin{cases} c_1(a - \theta) & \theta \leq a \\ c_2(\theta - a) & \theta > a \end{cases}$$

corresponds to a loss proportional to c_1 if we overestimate, and c_2 if we underestimate. Solving this loss function leads to $\hat{a}$ such that

$$\Pr[\theta \leq \hat{a}|y] = \frac{c_2}{c_1 + c_2} = \frac{c_2/c_1}{1 + c_2/c_1}$$

which presents an optimal quantile.

Example: Derive the posterior mean and posterior quantiles as the solution to quadratic and linear loss.
Quadratic Loss: $L(\theta, a) = (\theta - a)^2$

$$\begin{aligned}
\overline{L}(a) &= \int_\theta (\theta - a)^2 p(\theta|y) d\theta \\
&= \int_\theta \theta^2 p(\theta|y) d\theta - 2a \int_\theta \theta p(\theta|y) d\theta + a^2 \int_\theta p(\theta|y) d\theta \\
&= \int_\theta \theta^2 p(\theta|y) d\theta - 2aE[\theta|y] + a^2 \\
\implies 0 &= \frac{\partial}{\partial a} \overline{L}(a) \\
&= 0 - 2E[\theta|y] + 2\hat{a} \\
\hat{a} &= E[\theta|y].
\end{aligned}$$

Linear Loss: $L(\theta, a) = \begin{cases} c_1(a - \theta) & \theta \leq a \\ c_2(\theta - a) & \theta > a \end{cases}$

$$\begin{aligned}
\overline{L}(a) &= \int_{-\infty}^{a} c_1(a - \theta) p(\theta|y) d\theta + \int_a^\infty c_2(\theta - a) p(\theta|y) d\theta \\
\overline{L}(a) &= c_1 a \int_{-\infty}^{a} p(\theta|y) d\theta - c_1 \int_{-\infty}^{a} \theta p(\theta|y) d\theta + c_2 \int_a^\infty \theta p(\theta|y) d\theta - c_2 a \int_a^\infty p(\theta|y) d\theta \\
\overline{L}(a) &= c_1 a \mathrm{Pr}[\theta \leq a|y] - c_1 \int_{-\infty}^{a} \theta p(\theta|y) d\theta + c_2 \int_a^\infty \theta p(\theta|y) d\theta - c_2 a \mathrm{Pr}[\theta \geq a|y]
\end{aligned}$$

recall:

$$\frac{d}{dt} \int_{g(t)}^{f(t)} h(x) dx = h(f(t)) \cdot f'(t) - h(g(t)) \cdot g'(t).$$

Hence,

$$\begin{aligned}
\frac{d}{da} \overline{L}(a) &= c_1 \mathrm{Pr}[\theta \leq a|y] - \underbrace{a c_1 [\mathrm{Pr}(a|y) \cdot 1]}_{\mathrm{Pr}(a|y) = 0} - \underbrace{a c_2 [\mathrm{Pr}(-a|y) \cdot 1]}_{\mathrm{Pr}(-a|y) = 0} - c_2 \mathrm{Pr}[\theta \geq a|y] \\
0 &= c_1 \mathrm{Pr}[\theta \leq \hat{a}|y] - c_2 \mathrm{Pr}[\theta \geq \hat{a}|y] \\
0 &= c_1 \mathrm{Pr}[\theta \leq \hat{a}|y] + c_2 \mathrm{Pr}[\theta \leq \hat{a}|y] - c_2 \\
\mathrm{Pr}[\theta \leq \hat{a}|y] &= \frac{c_2}{c_1 + c_2}.
\end{aligned}$$

2.1.2 Interval Estimation: *credible* intervals

A $100 \times p\%$ equi-tailed *credible interval* $(0 < p < 1)$ is provided by

$$[\theta_i(\{1 - p\}/2), \theta_i(\{1 + p\}/2)]$$

2.2 Asymptotic Properties of Bayesian Estimators

In a situation where the data is i.i.d. we denote the posterior mean by $\tilde{\theta}_n = \tilde{\theta}_n(Y_n) = E[\theta|Y_n]$ and the MLE by $\hat{\theta}_n$. Then,

$$\sqrt{n}(\tilde{\theta}_n - \theta) = \sqrt{n}(\tilde{\theta}_n - \hat{\theta}_n) + \sqrt{n}(\hat{\theta}_n - \theta)$$

We know from the CLT that $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N_p(0, I_1^{-1}(\theta))$ where $I(\theta)$ is the information in a sample of size 1. It can be shown that $\sqrt{n}(\tilde{\theta}_n - \hat{\theta}_n) \xrightarrow{p} 0$ and so we have that,

$$\sqrt{n}(\tilde{\theta}_n - \theta) \xrightarrow{d} N_p(0, I_1^{-1}(\theta))$$

Note that the value of the prior diminishes as $n \to \infty$.

2.3 Prior Choice

2.3.1 Types of Priors

In Bayesian inference the choice of prior is a crucial aspect. There are two types of prior specification, **baseline** and **substantive**. A baseline prior had minimal impact and is used when we begin an approach. A quote from the textbook describes the substantive prior well,

> "An analysis with a baseline prior may be the only analysis performed or, alternatively, may provide an analysis with which other analyses in which substantive priors are specified may be compared. Such substantive priors constitute the second type of specification in which the incorporation of contextual information is required. Once we have a candidate substantive prior, it is often beneficial to simulate hypothetical data sets from the prior and examine these realizations to see if they conform to what is desirable. A popular label for analyses for which the priors are, at least in part, based on subject matter information is subjective Bayes."

2.4 Baseline Priors

Simply we may want to take the baseline prior to be 1, however this fails for two reasons. (1) The posterior may have an improper integration.

Example: Suppose $Y|p \sim \text{Binom}(n, p)$ where the focus is the logit, $\theta = \log[p/(1-p)]$. Then, if $\pi(\theta) \propto 1$, we have

$$\pi(p) \propto [p(1-p)]^{-1}$$

which is improper and can easily be seen as the posterior for p would be,

$$P(p|\boldsymbol{y}) \propto \binom{n}{y} p^y (1-p)^{n-y} \times [p(1-p)]^{-1}.$$

This is an improper prior because the resulting posterior (above) diverges upon integration. Another example can be seen on page 91-92 of the textbook.

(2) The second difficulty arrives when we reparameterize the model in terms of $\phi = g(\theta)$, where $g(\cdot)$ is a one-to-one mapping, then the prior for ϕ corresponds to

$$\pi(\phi) = \left| \frac{d\theta}{d\phi} \right|$$

From above example we had that $\theta = \log[p/(1-p)]$ and we were told that if $\pi(\theta) = 1$, then this implied that $\pi(p) \propto [p(1-p)]^{-1}$.

$$\pi(\theta) \propto 1 \implies \pi(p) \propto \left|\frac{d\theta}{dp}\right| = \left|\frac{d}{dp}\log[p/(1-p)]\right| = \left|\frac{1-p}{p} \cdot \frac{(1-p)+p}{(1-p)^2}\right| = [p(1-p)]^{-1}$$

For another example of this, consider a variance σ^2 where $\pi(\sigma^2) \propto 1$. From this we have,

$$\pi(\sigma^2) \propto 1 \implies \pi(\sigma) \propto \left|\frac{d}{d\sigma}\sigma^2\right| \propto \sigma$$

This is also improper because it's saying that the prior variance was non-constant.

2.4.1 Jeffery's Prior

Jefferys suggested use of

$$\pi(\theta) \propto |I(\theta)|^{1/2}$$

where $I(\theta)$ is the expected information. This prior has the desirable property of invariance to reparameterization. Consider $\phi = g(\theta)$ for the univariate case,

$$I(\phi) = I(\theta) \times \left(\frac{d\theta}{d\phi}\right)^2$$

This is completely glossed over in the book and there isn't a very good explanation as to what is actually going on. Sorry! However, if we start with

$$\pi(\phi) \propto |I(\phi)|^{1/2}$$

and we know $\phi = g(\theta)$, then

$$\pi(\theta) \propto I\left(g^{-1}(\phi)\right)^{1/2} \times \left|\frac{d\phi}{d\theta}\right|$$

Example Suppose $Y|p \sim \text{Binom}(n, p)$ where the focus is the logit, $\theta = \log[p/(1-p)]$. Then, Jeffery's prior in this case, where $I(p) = n/[p(1-p)]$, is

$$\pi(p) \propto [p(1-p)]^{-1/2}.$$

This is no longer improper, as the integral of the posterior resulting from this prior does not diverge.

2.5 Substantive Priors

substantive priors normally are described by a distribution so there really isn't a lot to talk about. Examples of using substantive priors will be done in the conjugacy question.

2.6 Model Mixtures

In some cases multiple models can be used to formulate a posterior. Say we have J many models, $\{M_1, \ldots, M_J\}$, each with some probability of belief, $\Pr(M_j)$, in which the parameters from that model have some belief, $\Pr(\boldsymbol{\theta}_j|M_j)$. We can represent the posterior based on all of the models with the following equations.

$$p(\phi|\mathbf{y}) = \sum_{j=1}^{J} p(\phi|M_j, \mathbf{y})\Pr(M_j|\mathbf{y})$$

where

$$\begin{aligned}
p(\phi|M_j, \mathbf{y}) &= \int p(\phi|\boldsymbol{\theta}_j, M_j, \mathbf{y})p(\boldsymbol{\theta}_j|M_j, \mathbf{y})d\boldsymbol{\theta}_j \\
&= \frac{1}{p(\mathbf{y}|M_j)} \int p(\phi|\boldsymbol{\theta}_j, M_j, \mathbf{y})p(\mathbf{y}|\boldsymbol{\theta}_j, M_j)p(\boldsymbol{\theta}_j|M_j)d\boldsymbol{\theta}_j \\
\Pr(M_j|\mathbf{y}) &= \frac{p(\mathbf{y}|M_j)\Pr(M_j)}{p(\mathbf{y})} \\
&= \frac{\int p(\mathbf{y}|\theta_j, M_j)p(\theta_j|M_j)d\theta_j \Pr(M_j)}{p(\mathbf{y})}.
\end{aligned}$$

To summarize for the posterior for ϕ, we might report the posterior mean

$$E[\phi|y] = \sum_{j=1}^{J} E[\phi|\mathbf{y}, M_j] \times \Pr(M_j|\mathbf{y})$$

which is simply the average of the posterior means across models, weighted by the posterior weight received by each model. The posterior variance is

$$\begin{aligned}
\mathrm{var}(\phi|\mathbf{y}) = &\sum_{j=1}^{J} \mathrm{var}[\phi|\mathbf{y}, M_j] \times \Pr(M_j|\mathbf{y}) \\
&+ \sum_{j=1}^{J} \{E[\phi|\mathbf{y}, M_j] - E[\phi|\mathbf{y}]\}^2 \times \Pr(M_j|\mathbf{y})
\end{aligned}$$

2.7 Hypothesis Testing

The posterior probability of H_0 given H_0 or H_1 is true is also known as the "relative truth" of H_0. While H_0 may provide a bad fit for the model, H_1 may be worse.

$$\Pr(H_0|\mathbf{y}, H_0 \cup H_1) = \frac{p(\mathbf{y}|H_0)\Pr(H_0|H_0 \cup H_1)}{p(\mathbf{y}|H_0 \cup H_1)}$$

If we wish to compare models H_0 and H_1, then a natural measure is given by the *posterior odds*

$$\frac{\Pr(H_0|\mathbf{y})}{\Pr(H_1|\mathbf{y})} = \frac{p(\mathbf{y}|H_0)}{p(\mathbf{y}|H_1)} \times \frac{\Pr(H_0)}{\Pr(H_1)}$$

where the *Bayes factor*

$$BF = \frac{p(\mathbf{y}|H_0)}{p(\mathbf{y}|H_1)}$$

is the ratio of the marginal distributions of the data under the two models, and $\Pr(H_0)/\Pr(H_1)$ is the *prior odds*. Care is required in the choice of priors when Bayes factors are calculated.

2.8 Implementation

2.8.1 Conjugacy

Posterior and *conjugate* priors have the same distributional form. Conjugate priors are commonly used for computational purposes and have closed forms which make them easier to computer. The book offers a table of priors, likelihoods, and posteriors which is filled with typos. We will derive the posteriors from the likelihood and priors and correct the table.

Table 3.1 Derivations

The fixed table

Prior	Likelihood	Posterior
$\theta \sim N(m, v)$	$\bar{Y}\|\theta \sim N(\theta, \sigma^2/n)$ with σ^2 known	$\theta\|\boldsymbol{y} \sim N[w\bar{y} + (1-w)m, w\sigma^2/n]$ with $w = \frac{v}{(v+\sigma^2/n)}$
$\theta \sim Be(a, b)$	$Y\|\theta \sim \text{Bin}(n, \theta)$	$\theta\|\boldsymbol{y} \sim Be(a+y, b+n-y)$
$\theta \sim Ga(a, b)$	$Y\|\theta \sim \text{Poisson}(\theta)$	$\theta\|\boldsymbol{y} \sim Ga(a+y, b+1)$
$\theta \sim Ga(a, b)$	$Y\|\theta \sim \text{Exp}(\theta)$	$\theta\|\boldsymbol{y} \sim Ga(a+1, b+y)$

1. Normal prior and Normal likelihood

$$
\begin{aligned}
p(\theta|y) &\propto [\theta \sim N(m,v)] \times [\bar{Y}|\theta \sim N(\theta, \sigma^2/n)] \\
&\propto \exp\left(-\frac{(\theta-m)^2}{2v}\right) \times \exp\left(-\frac{(\bar{Y}-\theta)^2}{2[\sigma^2/n]}\right) \\
&\propto \exp\left(-\frac{(\theta-m)^2}{2v} - \frac{(\bar{Y}-\theta)^2}{2[\sigma^2/n]}\right) \\
&\propto \exp\left(-\frac{1}{2v[\sigma^2/n]}\left[(\sigma^2/n)(\theta-m)^2 + v(\bar{Y}-\theta)^2\right]\right) \\
&\propto \exp\left(-\frac{1}{2v[\sigma^2/n]}\left[(\sigma^2/n)(\theta^2-2m\theta) + v(-2\theta\bar{Y}+\theta^2)\right]\right) \\
&\propto \exp\left(-\frac{1}{2v[\sigma^2/n]}\left[(\sigma^2/n)\theta^2 - (\sigma^2/n)2m\theta - 2v\theta\bar{Y} + v\theta^2\right]\right) \\
&\propto \exp\left(-\frac{1}{2v[\sigma^2/n]}\left[\theta^2(\sigma^2/n+v) - 2\theta((\sigma^2/n)m + v\bar{Y})\right]\right) \\
&\propto \exp\left(-\frac{1}{2w[\sigma^2/n]}\left[\theta^2 - 2\theta\left(\frac{(\sigma^2/n)m + v\bar{Y}}{(\sigma^2/n)+v}\right)\right]\right) \\
&\propto \exp\left(-\frac{1}{2w[\sigma^2/n]}\left[\theta^2 - 2\theta\left(\frac{(\sigma^2/n)m}{(\sigma^2/n)+v} + w\bar{Y}\right)\right]\right) \\
&\propto \exp\left(-\frac{1}{2w[\sigma^2/n]}\left[\theta^2 - 2\theta\left(m\left[\frac{(\sigma^2/n)+v}{(\sigma^2/n)+v} - \frac{v}{(\sigma^2/n)+v}\right] + w\bar{Y}\right)\right]\right) \\
&\propto \exp\left(-\frac{1}{2w[\sigma^2/n]}\left[\theta^2 - 2\theta\left(m\left[1-w\right] + w\bar{Y}\right)\right]\right) \\
&\propto \exp\left(-\frac{\left[\theta - \left(m\left[1-w\right] + w\bar{Y}\right)\right]^2}{2w[\sigma^2/n]}\right) \sim N[w\bar{y} + (1-w)m, w\sigma^2/n]
\end{aligned}
$$

2. Beta prior and Binomial likelihood

$$
\begin{aligned}
p(\theta|y) &\propto [\theta \sim Be(a,b)] \times [Y|\theta \sim \text{Bin}(n,\theta)] \\
&\propto \text{B}(a,b)^{-1}\theta^{a-1}(1-\theta)^{b-1} \times \binom{n}{y}\theta^y(1-\theta)^{n-y} \\
&\propto \theta^{a+y-1}(1-\theta)^{n-y+b-1} \\
&\propto \text{B}(a+y, n-y+b)^{-1}\theta^{a+y-1}(1-\theta)^{n-y+b-1} \sim Be(a+y, b+n-y)
\end{aligned}
$$

3. Gamma prior and Poisson likelihood

$$
\begin{aligned}
p(\theta|y) &\propto [\theta \sim Ga(a,b)] \times [Y|\theta \sim \text{Poisson}(\theta)] \\
&\propto \frac{b^a}{\Gamma(a)}\theta^{a-1}e^{-b\theta} \times \frac{\theta^y e^{-\theta}}{y!} \\
&\propto \theta^{a+y-1}e^{-\theta(b+1)} \\
&\propto \frac{(b+1)^{a+y}}{\Gamma(a+y)}\theta^{a+y-1}e^{-\theta(b+1)} \sim Ga(a+y, b+1)
\end{aligned}
$$

4. Gamma prior and Exponential likelihood

$$\begin{aligned} p(\theta|y) &\propto [\theta \sim Ga(a,b)] \times [Y|\theta \sim \text{Exp}(\theta)] \\ &\propto \frac{b^a}{\Gamma(a)}\theta^{a-1}e^{-b\theta} \times \theta e^{-\theta y} \\ &\propto \theta^a e^{-(b+y)\theta} \\ &\propto \frac{(b+y)^{a+1}}{\Gamma(a+1)}\theta^a e^{-(b+y)\theta} \sim Ga(a+1, b+y) \end{aligned}$$

2.8.2 Importance Sampling

Integrals can be expressed as expectations; for example consider

$$I = \int_0^1 f(\theta)d\theta = E_U[f(\theta)]$$

where the expectation is with respect to the uniform distribution $U(0,1)$. This formulation suggests the obvious estimator

$$\hat{I}_m = \frac{1}{m}\sum_{i=1}^m f(\theta^{(t)}),$$

with $\theta^{(t)} \sim_{iid} U(0,1)$, $t = 1, \ldots, m$. By the central limit theorem,

$$\sqrt{m}(\hat{I}_m - T) \xrightarrow{d} N(0, var(f)),$$

where $var(f) = E[f(\theta)^2] - I^2$ and we have assumed the latter exists. The form of the variance reveals that the efficiency of the method is determined by how variable the function f is, with respect to the uniform distribution over $[0,1]$. If f were constant, we would have zero variance!

To achieve an approximately constant function, we can trivially rewrite the integral as,

$$I = \int f(\theta)d\theta = \int \frac{f(\theta)}{g(\theta)}g(\theta)d\theta = E_g\left[\frac{f(\theta)}{g(\theta)}\right]$$

where we no longer restrict θ to lie in $(0,1)$. Define the estimator

$$\hat{I}_m = \frac{1}{m}\sum_{i=1}^m \frac{f(\theta^{(t)})}{g(\theta^{(t)})},$$

where $\theta^{(t)} \sim_{iid} g(\cdot)$, with

$$\sqrt{m}(\hat{I}_m - T) \xrightarrow{d} N(0, var(f/g)),$$

and

$$var(f/g) = E_g\left[\left(\frac{f}{g}\right)^2\right] - I^2$$

The latter may be estimated by

$$\hat{var}(f/g) = \frac{1}{m}\sum_{i=1}^m \left(\frac{f(\theta^{(t)})}{g(\theta^{(t)})}\right)^2 - \hat{I}_m^2.$$

The more samples we take the more accurate our interpretation will be (as $n \to \infty$). The uncertainty can be measured with a confidence interval that takes the form,

$$\left[\hat{I}_m - |z_{\alpha/2}|\frac{\sqrt{v\hat{a}r(f/g)}}{\sqrt{m}}, \hat{I}_m + |z_{\alpha/2}|\frac{\sqrt{v\hat{a}r(f/g)}}{\sqrt{m}}\right]$$

The accuracy also depends on g, we choose g so that g dominates the tails so that f/g is finite. We also would like g to computationally efficient to generate samples from. Lastly we would like g to mimic the curvature of f.

2.8.3 Direct sampling with conjugacy

This entire section is pulled from the book verbatim.
Suppose we have generated independent samples $\{\theta^{(t)}, t = 1, \ldots, m\}$ from $p(\theta|y)$, with $\theta^{(t)} = [\theta_1^{(t)}, \ldots \theta_p^{(t)}]$; we describe how much samples may be used for inference. The univariate marginal posterior for $p(\theta_j|y)$ may be approximated by the histogram constructed form the points $\theta_j^{(t)},\ t = 1, \ldots, m$. Posterior means $\hat{E}[\theta_j|y]$ may be approximated by

$$\hat{E}[\theta_j|y] = \frac{1}{m}\sum_{t=1}^{m}\theta_j^{(t)}$$

with other moments following in an obvious fashion. Coverage probabilities of the form $\Pr(a < \theta_j < b|y)$ are estimated by

$$\hat{\Pr}(a < \theta_j < b|y) = \frac{1}{m}\sum_{t=1}^{m} I(a < \theta_j^{(t)} < b),$$

where $I(\cdot)$ is the indicator function. We discuss how to estimate the standard error associated with the estimate

$$\hat{\mu_m} = \frac{1}{m}\sum_{t=1}^{m}\theta^{(t)}$$

of $\mu = E[\theta|y]$. By strong law of large numbers $\hat{\mu_m} \to \mu$ as $m \to \infty$, and the CLT gives

$$\sqrt{m}(\hat{\mu_m} - \mu) \xrightarrow{d} N(0, \sigma^2)$$

where $\sigma^2 = \text{var}(\theta|y)$ (assuming this variance exists). The Monte Carlo standard error is $\sigma/\sqrt{m}$, with consistent estimator of σ:

$$\hat{\sigma_m} = \sqrt{\frac{1}{m}\sum_{t=1}^{m}(g(\theta^{(t)}) - \hat{\mu_m})^2}.$$

By Slutsky's theorem

$$\frac{\hat{\mu_m} - \mu}{\hat{\sigma_m}/\sqrt{m}} \xrightarrow{d} N(0, 1)$$

as $m \to \infty$. An asymptotic confidence interval for μ is therefore,

$$\hat{\mu_m} \pm 1.96 \times \frac{\hat{\sigma_m}}{\sqrt{m}}$$

We may also obtain standard errors for functions that are not simple expectations. For example, consider the posterior variance of a univariate parameter θ.

$$\sigma^2 = \text{var}(\theta|y) = E[(\theta - \mu)^2|y]$$

where $\mu = E[\theta|y]$. An obvious estimator is

$$\hat{\sigma}_m^2 = \frac{1}{m}\sum_{t=1}^{m}(\theta^{(t)} - \hat{\mu_m})^2$$

The joint estimation of μ and σ^2 is then,

$$\sqrt{m}\left(\begin{bmatrix}\hat{\mu_m} \\ \hat{\sigma_m}^2\end{bmatrix} - \begin{bmatrix}\mu \\ \sigma^2\end{bmatrix}\right) \xrightarrow{d} N_2\left(\begin{bmatrix}0 \\ 0\end{bmatrix}, \begin{bmatrix}\sigma^2 & \mu_3^* \\ \mu_3^* & \mu_4^* - \sigma^4\end{bmatrix}\right)$$

where $\mu_j^* = E[(\theta - \mu)^j|y]$ is the jth central moment. The standard error of $\hat{\sigma}^2$ is estimated by

$$\sqrt{\frac{\hat{\mu}_{4,m}^* - \hat{\sigma}_m^4}{m}}$$

where $\hat{\mu}_{4,m}^* = \frac{1}{m}\sum_{t=1}^{m}(\theta^{(t)} - \hat{\mu}_m)^4$. This is highly unstable, unfortunately. Thus we need far larger sample sizes.

2.8.4 Direct sample using the rejection algorithm

In *rejection sampling* we wish to sample from distribution f where f can be represented as

$$f(x) = \frac{f^*(x)}{\int f^*(x)dx}$$

and we have some proposal distribution $g(\cdot)$ for which

$$M = \sup_x \frac{f^*(x)}{g(x)} < \infty$$

The rejection algorithm is quite simple actually, here is the outline with explanations.

(1) Generate $U \sim U(0,1)$. As well, independently generate $X \sim g(\cdot)$.

(1.e) We generate two random variables, one as uniform, and another from some distribution, g, that dominates f in the tails. Ideally we want the generation of X from g to be computationally efficient as well.

(2) Accept X if

$$U < \frac{f^*(X)}{Mg(X)}$$

otherwise return to 1.

(2.e) Here we are taking the value of X we generated, plugging it into the function f^* and g, then comparing f^*, g, and M in a fraction to the uniform number. If we plug in for M it's easy to see the fraction will always be within $(0, 1)$.

$$U < \frac{f^*(X)}{Mg(X)} = \frac{\frac{f^*(X)}{g(X)}}{\sup\limits_x \frac{f^*(x)}{g(x)}} = \frac{\text{random value}}{\text{largest possibility of that random value}} \in (0, 1)$$

The chance that X is accepted as a proper sample can be calculated as the *probability of acceptance* and is

$$\text{Pr(acceptance)} = \int_{-\infty}^{\infty} \frac{f^*(y)}{M} dy = p_a$$

In practice for generating samples from the posterior we can use a simplified method without all the integration. Assume we are working on generating samples from a posterior to estimate a parameter. Let θ denote the unknown parameter we wish to estimate, and assume we can evaluate the maximized likelihood as

$$M = \sup_\theta p(y|\theta) = p(y|\hat{\theta})$$

where $\hat{\theta}$ is the MLE. The algorithm proceeds as follows:

1. Generate $U \sim U(0, 1)$ and independently, sample from the prior $\theta \sim \pi(\theta)$.
2. Accept θ if

$$U < \frac{p(y|\theta)}{M},$$

   otherwise return to 1.

The probability that a point is accepted can be calculated as

$$p_a = \frac{\int p(y|\theta)\pi(\theta)d\theta}{M} = \frac{p(y)}{M}.$$

This version of the algorithm is very easy to implement as the MLE is often a routine calculation. Rejection sampling from the prior is quite intuitive. We can actually use the empirical rejection rate to estimate the normalizing constant instead of finding it out via derivation:

$$\tilde{p}(y) = \hat{p}_a \times M.$$

2.9 Gibbs Sampler

The book explains this well.

2.10 Metropolis Hasting Algorithm

The book explains this well.

3 Linear Models Inference

3.1 Model Specification

A multiple linear regression model takes the form

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + +\beta_k x_{ik} + \epsilon_i$$

where we begin by assuming the error terms are uncorrelated with $E[\epsilon_i] = 0$ and $\text{var}(\epsilon_i) = \sigma^2$. Simple regression is when $k = 1$. In matrix form the above equation is

$$\boldsymbol{Y} = \boldsymbol{x}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where

$$\boldsymbol{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \quad \boldsymbol{x} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix} \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

with $E[\boldsymbol{\epsilon}] = 0$ and $\text{var}(\boldsymbol{\epsilon}) = \sigma^2 \boldsymbol{I}_n$. We will also sometimes write

$$Y_i = \boldsymbol{x}_i \boldsymbol{\beta} + \epsilon_i$$

where $\boldsymbol{x}_i = [1, x_{i1}, \ldots, x_{ik}]$ for $i = 1, \ldots, n$. In order to estimate the values of β for a given model, the following estimator is derived to minimize the least squares equation.

$$\begin{aligned}
\sum_{i=1}^{n} (Y_i - \boldsymbol{x}_i \boldsymbol{\beta})^2 &= (\boldsymbol{Y} - \boldsymbol{x}\boldsymbol{\beta})^T (\boldsymbol{Y} - \boldsymbol{x}\boldsymbol{\beta}) \\
\implies 0 &= \frac{\partial}{\partial \hat{\boldsymbol{\beta}}} (\boldsymbol{Y} - \boldsymbol{x}\hat{\boldsymbol{\beta}})^T (\boldsymbol{Y} - \boldsymbol{x}\hat{\boldsymbol{\beta}}) \\
&= \left[\frac{\partial}{\partial \hat{\boldsymbol{\beta}}} (\boldsymbol{x}\hat{\boldsymbol{\beta}})^T \right] (\boldsymbol{Y} - \boldsymbol{x}\hat{\boldsymbol{\beta}}) \\
&= \boldsymbol{x}^T (\boldsymbol{Y} - \boldsymbol{x}\hat{\boldsymbol{\beta}}) \\
\implies \hat{\boldsymbol{\beta}} &= (\boldsymbol{x}^T \boldsymbol{x})^{-1} \boldsymbol{x}^T \boldsymbol{Y}
\end{aligned}$$

The estimator above is *Ordinary Least Square Estimator (LSE)* for $\boldsymbol{\beta}$. It is the,

1. MLE when the errors are i.i.d. and follow the normal distribution.
2. Posterior mean when the prior is proportional to $1/\sigma^2$.
3. BLUE (best linear unbiased estimator) when the first moment is $\boldsymbol{x}\boldsymbol{\beta}$ and the errors are uncorrelated and have common variance (Gaussian assumption is not required).

The LSE is consistent and asymptotic Gaussian

$$(\boldsymbol{x}^T \boldsymbol{x})^{1/2} (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} \boldsymbol{N}_{k+1}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_{k+1})$$

In the method of generalized least squares, we assume that $E[\boldsymbol{Y}|\boldsymbol{x}] = \boldsymbol{x}\boldsymbol{\beta}$ and $\text{var}[\boldsymbol{Y}|\boldsymbol{x}] = \sigma^2\boldsymbol{V}$ where $\boldsymbol{V}$ is a known matrix (weighted least squares corresponds to a diagonal $\boldsymbol{V}$) and consider the function,

$$RSS_G(\boldsymbol{\beta}) = (\boldsymbol{Y} - \boldsymbol{x}\boldsymbol{\beta})^T\boldsymbol{V}^{-1}(\boldsymbol{Y} - \boldsymbol{x}\boldsymbol{\beta})$$

Minimization of $RSS_G(\boldsymbol{\beta})$ yields the estimating function

$$G_G(\boldsymbol{\beta}) = \boldsymbol{x}^T\boldsymbol{V}^{-1}(\boldsymbol{Y} - \boldsymbol{x}\boldsymbol{\beta})$$

and estimator

$$\hat{\boldsymbol{\beta}}_G = (\boldsymbol{x}^T\boldsymbol{V}^{-1}\boldsymbol{x})^{-1}\boldsymbol{x}^T\boldsymbol{V}^{-1}\boldsymbol{Y}$$

with asymptotic distribution

$$(\boldsymbol{x}^T\boldsymbol{V}^{-1}\boldsymbol{x})^{1/2}(\hat{\boldsymbol{\beta}}_G - \boldsymbol{\beta}) \xrightarrow{d} \boldsymbol{N}_{k+1}(\boldsymbol{0}, \sigma^2\boldsymbol{I}_{k+1})$$

This estimator arises when $\boldsymbol{\epsilon} \sim \boldsymbol{N}_n(\boldsymbol{0}, \sigma^2\boldsymbol{V})$ with $\boldsymbol{V} = \boldsymbol{I}_n$ giving the OLSE as expected. An unbiased estimator of σ^2 is

$$\hat{\sigma}_G^2 = \frac{1}{n-k-1}(\boldsymbol{Y} - \boldsymbol{x}\hat{\boldsymbol{\beta}})^T\boldsymbol{V}^{-1}(\boldsymbol{Y} - \boldsymbol{x}\hat{\boldsymbol{\beta}})$$

3.2 Gauss-Markov Theorem

Consider a linear regression setup as we had before,

$$\boldsymbol{Y} = \boldsymbol{x}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where

$$\boldsymbol{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \quad \boldsymbol{x} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix} \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

with $E[\boldsymbol{Y}] = \boldsymbol{x}\boldsymbol{\beta}$ and $\text{cov}(\boldsymbol{Y}) = \sigma^2\boldsymbol{I}_n$. The Gauss-Markov Theorem states that, given the above setup, $\hat{\boldsymbol{\beta}} = (\boldsymbol{x}^T\boldsymbol{x})^{-1}\boldsymbol{x}^T\boldsymbol{Y}$ is the best linear unbiased estimator (BLUE) of $\boldsymbol{c}^T\boldsymbol{\beta}$.
A BLUE estimator of $\boldsymbol{\beta}$

1. Is a linear function of $\boldsymbol{Y}$, so that the estimator can be written $\boldsymbol{B}^T\boldsymbol{Y}$, for an $n \times k+1$ matrix $\boldsymbol{B}$.

2. Is unbiased so that $E[\boldsymbol{B}^T\boldsymbol{Y}] = \boldsymbol{\beta}$.

3. Has the smallest variance among all linear estimators.

The MLE is the LSE when the errors are uncorrelated and have common variance:
Consider the model above with parameter vector $\boldsymbol{\theta} = [\boldsymbol{\beta}, \sigma]$ and is of dimension $p \times 1$ where $p = k+1$. The likelihood function (which is found from the gaussian density) is

$$L(\boldsymbol{\theta}) = (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2}(\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta})^T(\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta})\right]$$

From here we can solve for the solutions of $\boldsymbol{\theta}$ to the maximization of the likelihood function. These solutions are

$$\begin{aligned}
\hat{\boldsymbol{\beta}} &= (\boldsymbol{x}^T\boldsymbol{x})^{-1}\boldsymbol{x}^T\boldsymbol{Y} \\
\hat{\sigma} &= \left[\frac{1}{n}(\boldsymbol{Y} - \boldsymbol{x}\hat{\boldsymbol{\beta}})^T(\boldsymbol{Y} - \boldsymbol{x}\hat{\boldsymbol{\beta}})\right]^{1/2}
\end{aligned}$$

The means of these estimators are trivially their statistics, $E[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$ and $E[\hat{\sigma}] = \sigma$. The variance can be found via algebra.

$$\begin{aligned}
\text{var}(\hat{\boldsymbol{\beta}}) &= \text{var}((\boldsymbol{x}^T\boldsymbol{x})^{-1}\boldsymbol{x}^T\boldsymbol{Y}) \\
&= ((\boldsymbol{x}^T\boldsymbol{x})^{-1}\boldsymbol{x}^T)\text{var}(\boldsymbol{Y})((\boldsymbol{x}^T\boldsymbol{x})^{-1}\boldsymbol{x}^T)^T \\
&= \sigma^2((\boldsymbol{x}^T\boldsymbol{x})^{-1}\boldsymbol{x}^T)((\boldsymbol{x}^T\boldsymbol{x})^{-1}\boldsymbol{x}^T)^T \\
&= \sigma^2(\boldsymbol{x}^T\boldsymbol{x})^{-1}\ \underbrace{\boldsymbol{x}^T\boldsymbol{x}(\boldsymbol{x}^T\boldsymbol{x})^{-1}}_{\text{The Identity Matrix}} \\
&= \sigma^2(\boldsymbol{x}^T\boldsymbol{x})^{-1}
\end{aligned}$$

Therefore,

$$\hat{\text{var}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2(\boldsymbol{x}^T\boldsymbol{x})^{-1}$$

Similarly it can be solved for σ (do on your own)

$$\text{var}(\hat{\sigma}) = \frac{\sigma^2}{2n} \qquad\qquad \hat{\text{var}}(\hat{\sigma}) = \frac{\hat{\sigma}^2}{2n}.$$

The asymptotic distribution of the MLE is

$$(\boldsymbol{x}^T\boldsymbol{x})^{1/2}(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}) \xrightarrow{d} \boldsymbol{N}_{k+1}(\boldsymbol{0}, \sigma^2\boldsymbol{I}_{k+1})$$

3.3 Inference

1. Inference for β

   (a) z-based (standard normal; Slutsky's theorem):
   When the variance is known, an asymptotic $100 \times (1-\alpha)\%$ confidence interval for β_j is

   $$\hat{\beta}_j \pm z_{\alpha/2} \times \hat{\text{s.e.}}(\hat{\beta}_j)$$

   where $z_{\alpha/n} = \Phi(\alpha/2)$.

   (b) t-based (sample variance is used):
   When the variance is unknown and a sample quantity must be used, an asymptotic $100 \times (1-\alpha)\%$ confidence interval for β_j is

   $$\hat{\beta}_j \pm t_{\alpha/2\,,\,n-k-1} \times \hat{\text{s.e.}}(\hat{\beta}_j)$$

where $t_{\alpha/2\,,\,n-k-1}$ is the $\alpha/2$ percentage point of a standard t random variable with $n-k-1$ degrees of freedom.
From this we can derive a hypothesis test using the t test (to test $H_0:\ \beta_j = h_0$).

$$T = \frac{\hat{\beta}_j - h_0}{S_j^{1/2}\hat{\sigma}} \sim T_{n-k-1}$$

where $S_j^{1/2}$ is the $(j,j)^{th}$ element of $(\boldsymbol{x}^T\boldsymbol{x})^{-1}$ and T_{n-k-1} denotes the univariate t distribution with $n-k-1$ degrees of freedom, location $\hat{\beta}_j$, and scale $S_j\hat{\sigma}^2$.
The joint confidence interval for $\boldsymbol{\beta}$ is the ellipsoid

$$(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^T\boldsymbol{x}^T\boldsymbol{x}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) \leq (k+1)s^2F_{k+1,n-k-1}(1-\alpha)$$

where $s^2 = RSS/(n-k-1)$ and $F_{k+1,n-k-1}(1-\alpha)$ is the $1-\alpha$ point of the F distribution with $k+1, n-k-1$ degrees of freedom.

(c) F-based (comparing two models):
Example: $H_0 : \beta_j = h_0 = 0$
We test the full model versus the partial model. Let $\boldsymbol{\beta}_{-j} = [\beta_0, \ldots, \beta_{j-1}, \beta_{j+1}, \ldots, \beta_k]$. Then,

$$F = \frac{FSS(\beta_j|\boldsymbol{\beta}_{-j})/1}{RSS(\boldsymbol{\beta})/(n-k-1)} = \frac{[RSS(\boldsymbol{\beta}_{-j}) - RSS(\boldsymbol{\beta})]/1}{RSS(\boldsymbol{\beta})/(n-k-1)}$$

and under H_0, $F \sim F_{1,n-k-1}$.

Example: $H_0 : \boldsymbol{\beta}_1$ unrestricted, $\boldsymbol{\beta}_2 = \mathbf{0}$; $H_1 : \boldsymbol{\beta} = [\boldsymbol{\beta}_1, \boldsymbol{\beta}_2] \neq [\boldsymbol{\beta}_1, \mathbf{0}]$
We test the full model versus the partial model. In this case the partial model has multiple β_j' s equal to 0. Let $\boldsymbol{\beta}_1 = [\beta_0, \ldots, \beta_p]$ and $\boldsymbol{\beta}_2 = [\beta_{p+1}, \ldots, \beta_k]$. Then,

$$F = \frac{FSS(\boldsymbol{\beta}_2|\boldsymbol{\beta}_1)/(k-p)}{RSS(\boldsymbol{\beta})/(n-k-1)} = \frac{[RSS(\boldsymbol{\beta}_1) - RSS(\boldsymbol{\beta})]/(k-p)}{RSS(\boldsymbol{\beta})/(n-k-1)}$$

and under H_0, $F \sim F_{k-p,n-k-1}$.

2. Inference for σ^2

(a) Scaled chi-square.

(b) Not robust to departures from the normality.
If we wish to make inference about σ^2 then we might be tempted to construct a confidence interval for σ^2 by leaning on $\epsilon_i|\sigma^2 \sim_{iid} N(0, \sigma^2)$. This leads to

$$\frac{\sum_{i=1}^n (Y_i - \boldsymbol{x}_i\boldsymbol{\beta})^2}{\sigma^2} \sim \chi^2_{n-k-1}$$

Intervals obtained in this manner are extremely non-robust to departures from normality.